# Grandfathering at Sale:
## School Assignment Rights and Housing Exchange

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**Abstract**

Grandfathering protects incumbents but can impede exchange when protection expires at sale. Beijing's 2018 school-assignment reform did this: retaining a pre-2019 title preserved eligibility under the prior single-school rule, whereas transfer made the buyer subject to announced multi-school assignment. Using 7,340 completed resales and historical assignments reconstructed from contemporaneous notices, I show that formerly key-school communities recorded roughly six sales per community before and after implementation, while ordinary-school communities rose from roughly three to five. A seasonal design yields a sharp exposed-relative implementation gap, but its magnitude is benchmark-sensitive. There is no detectable pre-deadline acceleration, and a separately benchmarked cumulative comparison detects no two-year differential. Among completed trades, price and discount point estimates move little while marketing-duration estimates rise. A search-and-bargaining model shows how a sale-triggered reset can remove marginal matches, make the selected mean price unsigned, and delay rather than destroy exchange. Comparing title-, property-, household-, and calendar-based protection under stated partial-equilibrium conditions makes a time-limited household entitlement one concrete benchmark to evaluate; the data do not rank transition rules in welfare terms.

**Keywords:** grandfathering; nonportable rights; school assignment; housing transactions; Beijing.
**JEL codes:** R21, R31, I28, D83.

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# 1 Introduction

Grandfathering protects people who relied on an old rule. It can also impede exchange when protection is conditional on keeping the asset. A homeowner may retain a below-market mortgage only by keeping the standing loan, a favorable property-tax basis only by remaining in the current home, or a public benefit only while holding an old title. Such rules create a general policy trade-off between incumbent protection and asset reallocation (Ferreira, 2010; Batzer et al., 2024; Fonseca and Liu, 2024). This paper asks how title-contingent grandfathering changes housing exchange and what object a transition right should follow.

The economic distinction is simple. If an amenity changes symmetrically for the incumbent and every prospective buyer, its value can be capitalized into price. If the incumbent preserves one entitlement by retaining the asset while a buyer receives a different entitlement after purchase, seller and buyer no longer value the same bundle. Price can divide the gains from trade that remain; it cannot transfer the incumbent's right. Selling extinguishes that right and, when the prior mapping is more valuable than the expected post-transfer bundle, raises the surplus required for a match. Because prices are then observed only for matches that clear the higher threshold, the mean price among completed transactions need not reveal the size or even the sign of the wedge.

I study this problem in Beijing's Haidian district. Residential addresses historically mapped eligible households to particular public primary schools. A reform announced in 2018 preserved the prior single-school rule for housing titles registered before January 1, 2019 while they remained under the old title. A buyer whose acquisition generated a post-cutoff title instead became subject to the announced multi-school rule (Haidian District Education Commission, 2018b,a). The physical dwelling did not change at the cutoff, but the assignment regime obtained through purchase did. The setting therefore supplies a legal date, a demarcated set of more exposed homes, and a record of exposure that can be reconstructed without using transaction outcomes.

The short-run quantity response alone is not the puzzle: a weaker buyer-side school bundle should reduce willingness to pay. The puzzle is its incidence and horizon. The title-vintage rule is permanent, yet the broad announcement-to-cutoff design detects no pre-deadline acceleration; a sharp relative completed-flow gap appears in the sixteen-week implementation comparison; and a separately benchmarked cumulative comparison detects no two-year differential. The paper asks whether a sale-triggered bilateral asymmetry, selection into completed trade, and finite-horizon clearance can organize that sequence.

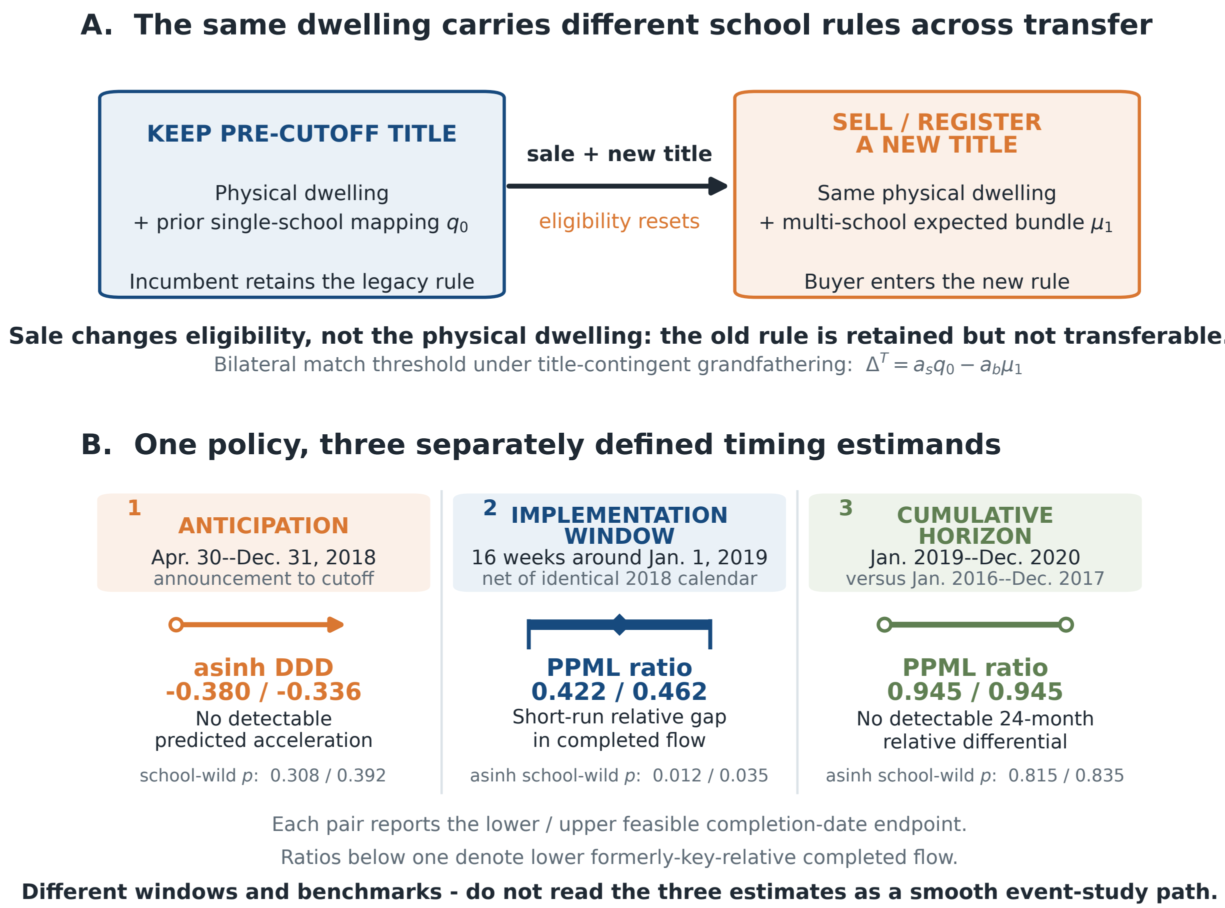


Figure 1: A sale-triggered rights reset and the timing trilemma

Notes: Panel A contrasts the bundle preserved by retaining a pre-cutoff title with the bundle obtained after transfer. Panel B reports separately defined exposed-relative completed-flow estimands at the two feasible completion-date endpoints: the broad announcement-to-cutoff acceleration test, the sixteen-week implementation comparison, and the cumulative twenty-four-month comparison. The paired estimates are not points on a smooth event-study path. Wild-cluster $p$-values refer to the corresponding asinh specifications; ratios are estimated separately by PPML.

Three facts build the answer. *First, formerly key-school communities miss the completed-sales rebound.* Depending on the feasible completion-date convention, their design-weighted counts move from 6.4 to 6.1 or from 6.0 to 6.5 transactions per community, while ordinary-school communities rise from roughly 2.9 to 4.9–5.0. The designated one-year seasonal comparison yields exposed-relative completed-flow ratios of 0.42–0.46; benchmarks that predate or average over the unusually positive 2018 differential preserve the sign but reduce the magnitude. The result is a failure to share the ordinary-community rebound, not an absolute collapse in exposed or districtwide transactions.

*Second, the divergence is concentrated near implementation.* The broad pre-deadline acceleration estimate is statistically indistinguishable from zero, the sixteen-week implementation contrast is sharp, and a separately benchmarked cumulative comparison yields a two-year ratio of 0.945 with no detectable differential. This timing trilemma rejects neither delay nor reallocation, but it is difficult to reconcile with permanent missing trade or a simple pull-forward story alone.

*Third, flow and selected time move more than the price of what trades.* Conditional on completion, price and discount point estimates move by about one percent while marketing-duration point estimates rise by 0.70–0.75 log points. Within listings known eventually to transact, selected entry has an exposed-relative ratio of 0.96 while exits reproduce ratios of 0.42–0.46; independent active-stock snapshots produce ratios of 0.83–0.92. The conditioning sets differ, so this configuration is consistent with delayed execution or reallocation without identifying a common listing hazard.

The design compares nineteen communities historically assigned to seven schools classified as key with forty ordinary-school communities. Historical assignment is reconstructed from contemporaneous admission notices. The preferred frame contains 7,340 completed transactions, including 1,173 in the four main sixteen-week cells. A seasonal triple difference compares formerly key and ordinary communities, before and after January 1, and the 2019 policy experiment with the identical 2018 calendar contrast. Missing day-level completion dates are bounded by a fixed one-day interval; assigned-school wild and CR2 inference, leave-outs, literal-notice controls, bandwidth and donut checks, and an outcome-blind spatial match are reported with the main estimates.

A compact search-and-bargaining model has four jobs. When the prior mapping is at least as valuable as the expected post-transfer bundle, the title-reset component raises the bilateral match threshold. A higher threshold has three parallel consequences: it removes marginal matches, slows clearance, and changes selection into the prices observed. In shorthand,

$$\text{title reset} \longrightarrow \text{higher threshold} \longrightarrow \begin{cases} \text{fewer accepted matches,} \\ \text{slower clearance,} \\ \text{selected-price response unsigned.} \end{cases}$$

The same primitives compare whether protection follows the title, property, household, or a common transition date. Under $q_0 \geq \mu_1$ and the other stated partial-equilibrium assumptions, each portable or uniform rule weakly expands the match set relative to title-contingent grandfathering. A time-limited household entitlement is one concrete benchmark worth evaluating when the objective is to protect incumbent reliance without making sale the forfeiture event; neither the model nor the data select it as optimal.

The paper contributes at the intersection of school capitalization, housing exchange, and nonportable-right lock-in. School access is well known to capitalize into property values (Black, 1999; Gibbons and Machin, 2003; Bayer et al., 2007; Fack and Grenet, 2010); recent Beijing studies document price, sorting, and spatial responses to multi-school zoning (Guan et al., 2024; Yang et al., 2024; Dai et al., 2025, 2026; Chen et al., 2026); and Tong et al. (2024) study price and enrollment-distribution effects of the same Haidian reform. School-quality changes have also been linked to selling time (Zahirovic-Herbert and Turnbull, 2008; Liu and Smith, 2023). The contribution is not that school policy can move price, completed volume, or marketing time separately.

The distinct empirical object is a sale-triggered change in the bundle being exchanged. The closest published transaction-volume evidence is Shao et al. (2023), who find announcement-period increases and enforcement-period declines after a later Xicheng reform. This paper follows the Haidian title-vintage clause through completed flow, the horizon of completion, and selected price and duration while keeping transaction exits, eventual-completer listings, and active stocks on separate risk sets.[1] The reduced form identifies an exposed-relative short-run exchange response; it does not identify a full-market sale hazard or uniquely separate match acceptance from demand, seller entry, withdrawal, reallocation, or platform coverage.

Conceptually, the paper connects the title rule to nonportable-benefit lock-in. Property-tax grandfathering and mortgage-rate lock-in show that favorable incumbent terms can suppress mobility (Ferreira, 2010; Fonseca and Liu, 2024; Batzer et al., 2024), while Hou et al. (2025) document housing-market responses when school-enrollment rights change for migrant families. The organizing object here is the attachment of a public right—to the title, property, household, or calendar. The model orders the resulting bilateral match sets under stated partial-equilibrium conditions. That transition-design comparison, rather than a welfare ranking, is the paper's conceptual payoff.

The remainder of the paper proceeds as follows. Section 2 describes the assignment system and title cutoff. Sections 3 and 4 present the observation universes and empirical design. Section 5 documents the three facts. Section 6 develops the model they motivate. Section 7 turns the model to transition design and external validity.

# 2 Institutional Setting

## 2.1 Address-based school access before the reform

Public primary-school access in Beijing has historically been tied closely to household registration and residence. In the single-school regime, an eligible address was mapped to one responsibility school. That mapping gave ownership an entitlement-like component: a household buying a home in the zone could use the address when applying for admission, subject to household-registration and occupancy requirements. A large empirical literature finds that school access capitalizes into housing values because residence and enrollment are jointly chosen (Black, 1999; Bayer et al., 2007; Fack and Grenet, 2010). In Beijing the institutional link is particularly visible in the category of "school-district housing."

The relevant treatment is not simply whether a home was located in Haidian. Every community in the estimation sample was in the district, and every one faced the reform. Former key-school status proxies for the value of the prior single-school mapping that was lost; it is not an observed household-specific treatment intensity. The treated schools are drawn from the published key-school list in Zheng et al. (2025). A community enters the treated group only when a visibly dated 2017 enrollment-notice image names it in the school's admission scope. Campus-brand, renamed-school, and loose affiliation links are excluded. Ordinary

[1] A completed HKBU project record (Zhang, 2022) describes planned analysis of a Beijing multi-school reform with assignment keyed to property-transaction timing and with price and transaction-volume dynamics. No public manuscript or findings were located as of August 2026, and the public description does not name the district.

controls come from the pre-reform platform school map, exclude every strict or affiliated key-school match, and are chosen by geography within Haidian.

## 2.2 The 2018 announcement and 2019 cutoff

The policy document was dated April 28, 2018 and published April 30. In the author's translation, its operative clause states: "Beginning January 1, 2019, houses newly registered and issued a real-property certificate will no longer correspond to one school; enrollment will follow multi-school zoning" (Haidian District Education Commission, 2018b). A December 2018 district notice repeated the same effective date and rule (Haidian District Education Commission, 2018a). The notice does not publish each address's post-reform school set, assignment probability, or capacity constraint. Hence the policy created a known title-vintage cutoff but not an observed continuous treatment dose.

An official district retrospective reports that computerized allocation for families who bought after the title cutoff was first used in 2021 (Haidian District, 2022). The January 2019 design therefore identifies market responses to the title rule and its expected assignment consequences, not responses to contemporaneous realized lottery draws.

The title-vintage clause is economically central. A pre-cutoff title remained under the prior single-school rule while held; a buyer whose acquisition generated a post-cutoff title entered the new assignment regime. Admission still depended on household-registration and occupancy requirements. The reform therefore did not simply lower a neighborhood amenity for everyone at once; it made the governing school-assignment rule conditional on whether the asset was transferred.

Panel A of Figure 1 makes the economic contrast explicit. If the incumbent keeps the dwelling, the physical house and the prior single-school mapping remain bundled. If the dwelling is sold, the physical asset passes to the buyer but the assignment regime resets. Price can divide the surplus from the house; it cannot make the buyer's post-transfer mapping identical to the incumbent's retained mapping. The empirical exposure measure asks where that reset was likely to matter most.

The economic prediction is ambiguous outside a narrow model. Before the cutoff, households could accelerate eligible transfers to preserve the prior single-school mapping. After the cutoff, the expected school bundle could fall, assignment uncertainty could rise, and buyers and sellers could take longer to locate an acceptable surplus. A broad grace-period surge is therefore one possible response, but not a logical necessity. The analysis treats that announcement-period response and the local enforcement response as distinct predictions; both were specified before reconstruction of the missing completion dates.

The cutoff is legally defined by registration and certification, not by the platform's transaction-completion timestamp. Administrative processing may separate the two. This distinction is why I use sixteen-week windows, show bandwidth and donut sensitivity, and avoid the language of unit-level sharp treatment assignment. The estimand is a differential completed-flow response around enforcement, not a local average treatment effect for homes one day apart in deed issuance.


Figure 2: Policy and identification timeline

Notes: The shaded regions are the sixteen-week estimation windows around January 1 in 2018 and 2019. January 1, 2018 supplies the identical-calendar seasonal comparison and is not a policy cutoff.

# 3 Data and Sample Construction

The observed outcome is housing exchange, not household migration: the archive does not link sellers or buyers across addresses.

The project combines three empirically distinct objects. First, the canonical platform archive contains completed resale transactions and supplies the main outcome. Second, posting and completion clocks within that archive construct a selected pipeline consisting only of listings known eventually to complete. Third, independent public crawler snapshots observe active offers at particular dates. Keeping these objects separate is not semantic housekeeping: it determines which denominators exist and which market stages can be studied.

## 3.1 Three observation universes

*Completed transactions.* The main archive records successful exits from the resale process. A month-only source completion date is still a completed transaction when the row carries a positive deal price and a marketing-duration clock. Missing a parsed day is therefore not evidence of non-sale. These records identify the timing and location of platform-recorded completions, but not the number of properties ever offered.

*The eventual-completer pipeline.* For completed records, posting dates reveal when each ultimately successful listing entered the platform pipeline. This creates exact entry, stock, and exit accounting within the selected set $I_i = 1$ of eventual completers. It is useful for locating the timing difference inside observed successful spells, but conditioning on $I_i = 1$ excludes withdrawals and censored listings by construction.

*Public active-stock snapshots.* Three independently collected public files contain genuine active listings: a November 2017 crawl with exact listing IDs (Huang, 2017), a late-March 2018 cross-section distributed with the H4M data (Zhao et al., 2022), and a broad February 2019 community crawl. These files establish that listing observations exist outside the completed archive. They do not share stable listing identifiers, timestamps, or crawler ledgers and therefore cannot be stacked into a property-level transition panel. I use them only on identical community supports and expose the cross-crawler comparability assumption through a sensitivity parameter.

## 3.2 Completed transactions

The transaction archive is a processed export from a proprietary listing platform covering Beijing resale records posted from 2010 through January 2021; the analysis period is 2015–2020. The canonical input contains 586,312 completed transactions after the project's cleaning screens. Every retained record has a positive transaction price, posting date, and marketing duration. Some observations lack a parsed day-level completion date because the source date is month-only, but that field does not indicate that the property failed to sell. The analysis therefore uses counts of platform-recorded completed transactions rather than a listing-level sale probability.

For the preferred notice-verified Haidian frame, the archive supplies 7,340 resolved transactions across fifty-nine communities: nineteen exposed communities and forty ordinary-school comparisons. Of these, 1,173 fall inside the four sixteen-week estimation windows. The regression contains 236 community–experiment–side observations. The online appendix summarizes the estimation samples and reports the archive-wide measurement audit.

### 3.3 Completion dates

Let $a_i$ be listing $i$'s posting date and $W_i$ its recorded marketing duration. When an exact completion date $d_i$ is present, I retain it. When it is missing, I use the fixed interval

$$d_i \in [a_i + W_i - 1,\ a_i + W_i]. \tag{1}$$

The lower endpoint follows the modal inclusive-day convention; the upper endpoint is the one-day sensitivity. The convention was fixed before re-estimating the policy coefficient.

The validation sample contains 414,979 transactions with both an exact date and the duration clock. Reconstructed and exact duration have correlation 0.99999, and all but one reconstructed date lies within one day of the reported date. In the broader sixty-one-community matched frame, 2,515 of 7,410 dates use the reconstruction; twenty-seven of those reconstructed dates (1.1 percent, or 0.36 percent of all rows) change cutoff-cell membership between endpoints. Both endpoints remain visible throughout the completed-flow analysis. The archive does not document platform market share, off-platform trades, or a stable coverage denominator; a treatment-correlated coverage change therefore remains possible.

### 3.4 Historical assignments and matched support

Historical exposure is fixed from 2017 school-admission notices. The exposed sample contains nineteen communities tied to seven schools externally classified as key. An outcome-blind archival reconstruction of the 2018 notice images verifies that all nineteen exposed communities remained assigned to the same seven schools one year later; this check neither selects units nor rematches the design. The images are retained with source pages and hashes, but come from a public secondary archive rather than an authenticated district database. Among the original ordinary-school comparisons, thirty-one communities appear literally in the 2017 notice text and nine are linked by an exact, unambiguous property or address alias. The preferred sample retains those forty controls; an exact-notice sensitivity retains only the thirty-one literal matches. Neither restriction rematches the treated communities.

Each exposed community is matched to as many as five nearby eligible ordinary-school communities, prioritizing the same pre-reform school district and otherwise remaining within Haidian. Reused controls receive inverse-reuse weights, normalized within the matched design. Inference clusters by historically assigned school. The preferred sample contains twenty-one school clusters, seven of which are exposed, so the headline analysis exhausts all $2^{21}$ assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure and also reports leverage-aware CR2/Satterthwaite inference (Pustejovsky and

Tipton, 2018) together with leave-one-out diagnostics. This enumeration removes Monte Carlo error but is not assignment-based randomization inference.

The external key-school classification is not created from the policy outcome. As an ex ante validation, I estimate a school component from 2015–2016 transaction prices and ask whether it predicts 2017 values before the announcement. The component lowers 2017 transaction-level mean squared error, its school-mean validation slope is 0.934, and formerly key schools have a value dose 0.851 standard deviations above ordinary schools. The online appendix reports the full diagnostic and design summary. A separately frozen continuous-dose policy specification fails its promotion gates, so the binary classification is retained as a transparent exposure ordering rather than replaced by an outcome-selected continuous treatment.

### 3.5 Outcome construction and aggregation

The main dependent variable is the number of platform-recorded completed transactions in each community–experiment–side cell. Each community appears four times: before and after January 1 in the 2018 seasonal experiment and before and after January 1 in the 2019 policy experiment. Zero-count cells are retained. The preferred linear specification applies the inverse hyperbolic sine to accommodate zeros without adding an arbitrary constant; PPML estimates the same saturated contrast on the count scale.

The sixteen-week window is long enough to accommodate search and the administrative separation between platform completion and title certification while remaining local to enforcement. I report the complete four- to sixteen-week bandwidth and donut family rather than choosing a window after reconstruction. Longer calendar-year and two-year outcomes test persistence, not the same local estimand.

For the timing analysis, marketing duration is the interval from posting to platform completion. A separate attention file is linked to the completed archive only through unique, outcome-blind exact matches. The frozen linked sample contains 855 eventual sales for physical showings and duration and 755 for page views. All attention outcomes are secondary because they condition on eventual completion; cumulative outcomes also mechanically load on exposure time.

## 4 Empirical Design

### 4.1 Estimating equation

Index communities by $c$, cutoff experiments by $y \in \{2018, 2019\}$, and sides of January 1 by $s \in \{\text{before}, \text{after}\}$. Each side contains sixteen weeks. Let $G_c$ indicate a formerly key-school community, $R_y$ the real 2019 cutoff experiment, and $P_s$ the post-January 1 side. For outcome $Y_{cys}$, the estimating equation is

$$\operatorname{asinh}(Y_{cys}) = \alpha_c + \lambda_{ys} + \rho(G_c \times R_y) + \phi(G_c \times P_s) + \tau_F(G_c \times R_y \times P_s) + \varepsilon_{cys}. \tag{2}$$

Community fixed effects absorb permanent level differences, and $\lambda_{ys}$ absorbs every experiment-by-side shock common to exposed and control communities. The lower-order exposure interactions allow the groups to differ across experiments and in average seasonality. The coefficient is

$$\tau_F = [\Delta_{A-B}(G = 1) - \Delta_{A-B}(G = 0)]_{2019} - [\Delta_{A-B}(G = 1) - \Delta_{A-B}(G = 0)]_{2018}, \tag{3}$$

where $\Delta_{A-B}$ is the after-minus-before change. A negative $\tau_F$ means that the exposed-control seasonal difference moved downward more around the 2019 cutoff than around the identical 2018 calendar date; it does not by itself imply that exposed counts fell in absolute terms. I also estimate the count model by PPML with the same design, weights, and fixed effects. Its reported relative post ratio is $\exp(\widehat{\tau}_F^{\mathrm{PPML}})$.

## 4.2 Identification and inference

The identifying restriction is parallel differential seasonality: absent the reform, the exposed-control difference in the sixteen-week January-to-spring change in 2019 would equal the corresponding difference in 2018. This restriction is more credible than a raw before-after comparison because housing activity is highly seasonal (Ngai and Tenreyro, 2014). It is less demanding than assuming exposed and control communities have equal levels or equal unconditional trends.

Five threats remain. First, another January 2019 shock could differentially affect precisely the seven key schools. Second, the processed archive could undergo a treatment-correlated coverage change. Third, the historical school map could be wrong. Fourth, ordinary-school communities could receive spillovers from the same districtwide reform. Fifth, the platform completion date may misalign with administrative certification. The design addresses, but cannot eliminate, these concerns through a prior-year calendar experiment, rolling false cutoffs, notice verification, both timing endpoints, local geographic controls, school-cluster inference, leave-one-community and leave-one-school estimates, and bandwidth/donut windows. For the fourth threat, Section 5.2 finds no detectable larger rebound among control communities nearest the exposed group, while leaving diffuse or delayed reallocation unresolved.

The control group is not untreated in the policy sense. Both groups entered multi-school zoning. The estimand contrasts replacement of a prior single-school mapping attached to a key school with replacement of an ordinary-school mapping. Raw 2019 counts are approximately flat in the exposed group and rebound sharply in the ordinary group, so the estimate may represent a suppressed counterfactual rebound in exposed communities, reallocation toward ordinary communities, or both. It is not the district-average effect. Spillovers that move ordinary controls in the same direction attenuate the differential; spillovers of the opposite sign can amplify it.

Before reconstructing missing completion dates, I recorded a broad announcement-period acceleration prediction as primary and the sixteen-week January comparison as a companion. The broad prediction is not supported, and the cutoff contrast does not satisfy every original promotion gate. An earlier exact-date screen had already revealed the local pattern, so the reconstructed-date and notice-restricted exercises validate a known signal rather than provide independent discovery. These timestamped internal protocols are not an external preregistration; the online appendix records the complete hierarchy and failed gates.

## 4.3 Outcome-blind spatial validation

A separately frozen design replaces the original many-to-one frame with a global maximum-cardinality, minimum-total-distance assignment that never uses transaction outcomes. It pairs all nineteen exposed communities without replacement to distinct notice-verified controls within 2.384 kilometers and gives each of the seven exposed schools equal weight. This is a local-counterfactual validation, not a boundary discontinuity or a second source of policy variation: the data contain community points rather than historical catchment polygons, and nearby controls may receive displaced transactions. The online appendix reports the pair map, construction, calipers, road-network and geographic-surface checks, Lunar-New-Year exclusions, and pre-policy diagnostics (Butts, 2023).

## 5 Three Facts about Housing Exchange

Three facts characterize the response to the title reset. Formerly key-school communities miss the ordinary-community rebound in completed exchange; the short-window divergence is not detectable in a separately benchmarked cumulative comparison; and quantity and selected duration move much more than prices among completed trades. The broad announcement-to-cutoff acceleration prediction is not supported at either completion-date endpoint. The surviving pattern is narrower and develops over eight to sixteen weeks, which is consistent with search and administrative processing rather than a literal one-day discontinuity.

### 5.1 Fact 1: Formerly key-school communities miss the rebound

Figure 3 shows the result before regression adjustment. In the preferred sample, 2019 design-weighted means move from 6.37 to 6.11 and from 6.00 to 6.47 in formerly key-school communities, compared with 2.91 to 4.87 and 2.79 to 4.99 in ordinary-school communities. The exposed counts therefore do not show a large observed contraction. In the identical-calendar 2018 comparison, both groups rise, with a larger increase in formerly key-school communities. The difference between these movements produces the relative regression result.

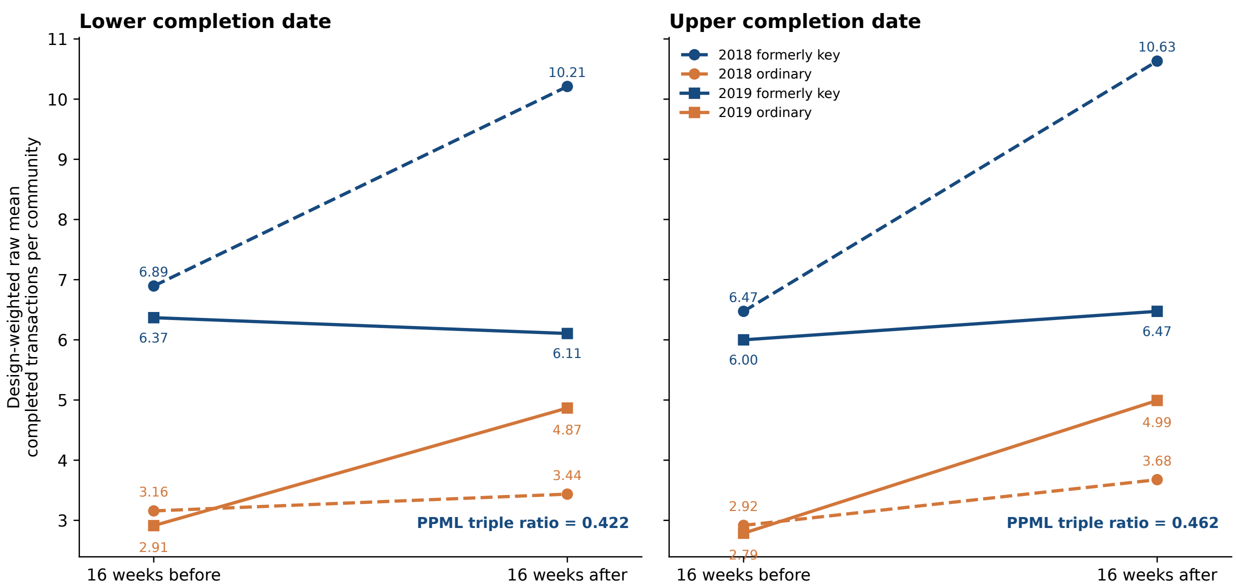


Figure 3: Formerly key-school communities miss the ordinary-community rebound

Notes: Each point is an inverse-reuse-weighted raw mean of completed transactions per community in a sixteen-week window. Color identifies formerly key-school and ordinary-school communities; dashed and solid lines identify the 2018 identical-calendar comparison and the 2019 policy experiment. The PPML triple ratio compares the 2019 formerly-key/ordinary after-before ratio with its 2018 analogue. It is a relative differential, not a districtwide volume ratio.

Table 1 reports the completed-flow DDD. After netting out 2018 differential seasonality, the asinh coefficients are −0.918 and −0.762, and separate PPML estimates imply ratios of 0.422 and 0.462. Under the one-year seasonal benchmark, these ratios represent exposed-relative gaps of 54–58 percent (Section 5.2 shows how the magnitude, though not the sign, depends on the benchmark year); they do not imply that exposed counts or total Haidian transactions fell by those percentages.

Exhaustively enumerated restricted-null school-wild tests for the asinh coefficients yield $p$-values of

0.012 and 0.035 across the two date conventions. CR2/Satterthwaite inference is more conservative at the upper endpoint, where the $p$-value is 0.062. I therefore emphasize agreement in sign and magnitude while retaining endpoint-specific uncertainty. The PPML ratios provide an interpretable scale; their conventional school-cluster tests are asymptotic and are not used for few-cluster significance claims. Literal notice-text controls, retained without rematching, imply similarly large relative gaps of 56.6 and 51.7 percent.

Table 1: Completed transaction flow around the January 2019 title-vintage cutoff

| | Lower completion date | Upper completion date |
|---|---|---|
| *Panel A. Preferred notice-verified sample* | | |
| Former key-school × After Jan. 1 × 2019 | $-0.918^{**}$ | $-0.762^{**}$ |
| School-clustered CR1 standard error | (0.335) | (0.376) |
| School-clustered CR1 $p$-value | [0.013] | [0.056] |
| Restricted-null school-wild $p$-value | [0.012] | [0.035] |
| CR2 standard error | (0.296) | (0.345) |
| CR2/Satterthwaite $p$-value | [0.017] | [0.062] |
| PPML relative completed-flow ratio | 0.422 | 0.462 |
| Asymptotic school-clustered PPML $p$-value | [0.043] | [0.037] |
| *Panel B. Literal notice-text controls only* | | |
| Asinh DDD | $-0.870^{**}$ | $-0.715^{*}$ |
| Restricted-null school-wild $p$-value | [0.022] | [0.056] |
| CR2/Satterthwaite $p$-value | [0.025] | [0.082] |
| PPML relative completed-flow ratio | 0.434 | 0.483 |
| *Panel C. Identifying support* | | |
| Completed transactions (preferred / literal-notice) | 1,173 / 1,029 | 1,173 / 1,029 |
| Communities (formerly key / preferred ordinary / literal ordinary) | 19 / 40 / 31 | 19 / 40 / 31 |
| Historically assigned schools (exposed / all) | 7 / 21 | 7 / 21 |
| Community–experiment–side cells (preferred / literal-notice) | 236 / 200 | 236 / 200 |

Notes: Each side contains sixteen weeks. The table reports the coefficient on Former key-school × After Jan. 1 × 2019 from an asinh DDD with community fixed effects, experiment-by-side fixed effects, lower-order exposure interactions, and inverse-reuse matching weights. A negative coefficient means the formerly-key-minus-ordinary after-before contrast moved downward more in 2019 than in the identical 2018 calendar experiment; it does not require formerly key-school counts to fall in levels. Stars apply only to asinh coefficients and use the displayed restricted-null wild $p$-values, which exhaust all $2^{21}$ assigned-school Rademacher sign vectors under the fully studentized procedure: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. These values are enumeration-exact under the bootstrap law, not assignment-based randomization inference. PPML ratios are scaling estimates; their displayed tests use asymptotic school-cluster covariance and carry no stars. Panel B changes only assignment provenance and does not rematch exposed communities. Raw weighted means appear in Figure 3.

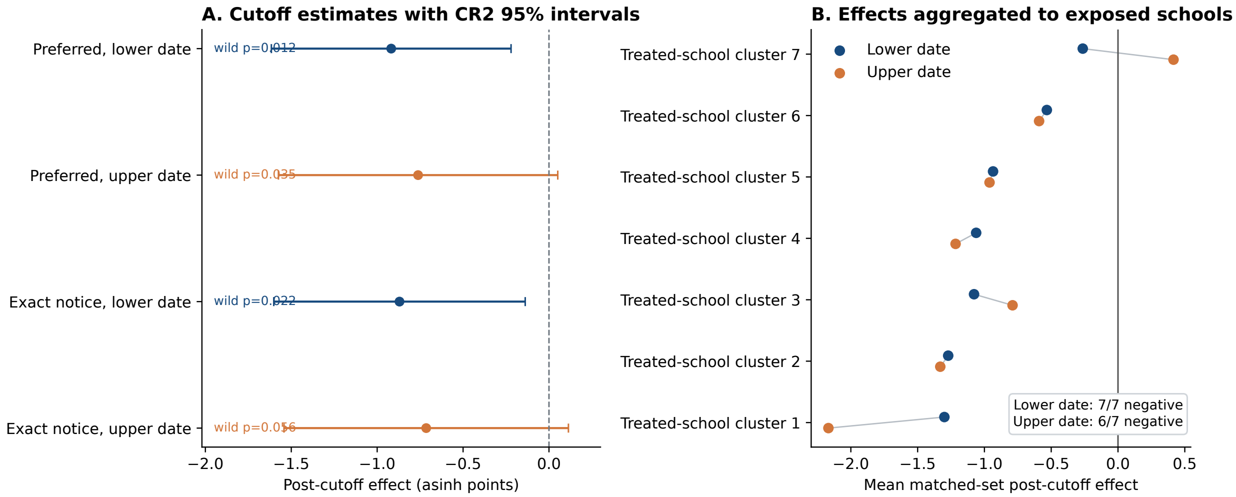


Figure 4: Cutoff estimates and exposed-school concentration

Notes: Panel A reports coefficient-specific CR2/Satterthwaite 95 percent intervals and exhaustively enumerated restricted-null wild-cluster $p$-values. Panel B averages matched-set effects within the seven exposed schools. Sign counts describe concentration and are not randomization inference.

## 5.2 What identifies the first fact

Leaving out any original community or assigned school without rematching preserves a negative DDD coefficient. In the preferred sample, the lower-date estimates range from −1.146 to −0.750 and the upper-date estimates range from −0.994 to −0.595. Sixteen of nineteen matched-set effects are negative under either date convention, and the school averages are negative for seven of seven exposed schools at the lower date and six of seven at the upper date. All ten specifications in the bandwidth/donut family fixed before date reconstruction produce negative estimates, but only eleven of the twenty endpoint-specific tests have wild $p < 0.05$. January 2019 ranks first and second in absolute magnitude among fourteen dates at the lower and upper endpoints. The online appendix displays the complete bandwidth, donut, and pseudo-cutoff family without significance filtering.

The outcome-blind spatial match produces a school-equal estimate of −0.72 at either date convention. Assumption-based seven-school sign-flip $p$-values are 0.047 and 0.078; six of seven school effects are negative, and every exposed-school omission preserves the sign. The estimates remain negative across the 2.0–3.0 kilometer calipers, but full support begins at 2.5 kilometers and the 1.5 kilometer match is near zero. Linear geographic surfaces preserve the headline result; quadratic surfaces preserve its magnitude but not conventional significance. These checks reduce concern that the result compares distant parts of Haidian, but they do not remove the one-date or differential-seasonality assumptions.

The most consequential diagnostic is the choice of seasonal benchmark. Figure 5 plots the formerly-key-minus-ordinary winter-to-spring gap using weights selected only from January 2015–August 2016 flows and coordinates. The 2016 point is near zero, 2017 is negative, and 2018 is unusually positive. Benchmarking 2019 against that 2018 differential therefore produces the largest estimate, −0.85 to −0.96 in this fixed-weight frame. The fully held-out 2018-versus-2017 contrast is itself sizeable and opposite-signed (0.61 and 0.63; wild $p$ = 0.068 and 0.049). Benchmarks that predate or average over the oscillation yield smaller estimates of −0.24 to −0.64, while the raw 2019 gap is −0.39 before subtracting any benchmark. The multiyear comparisons were added after observing the adverse placebo and remain labeled post-protocol. Every comparison preserves the negative sign; the credible magnitude is benchmark-sensitive. The online appendix discusses the coincidence of the 2017 movement with Beijing's March 2017 credit tightening, but that timing is not used to validate the design.

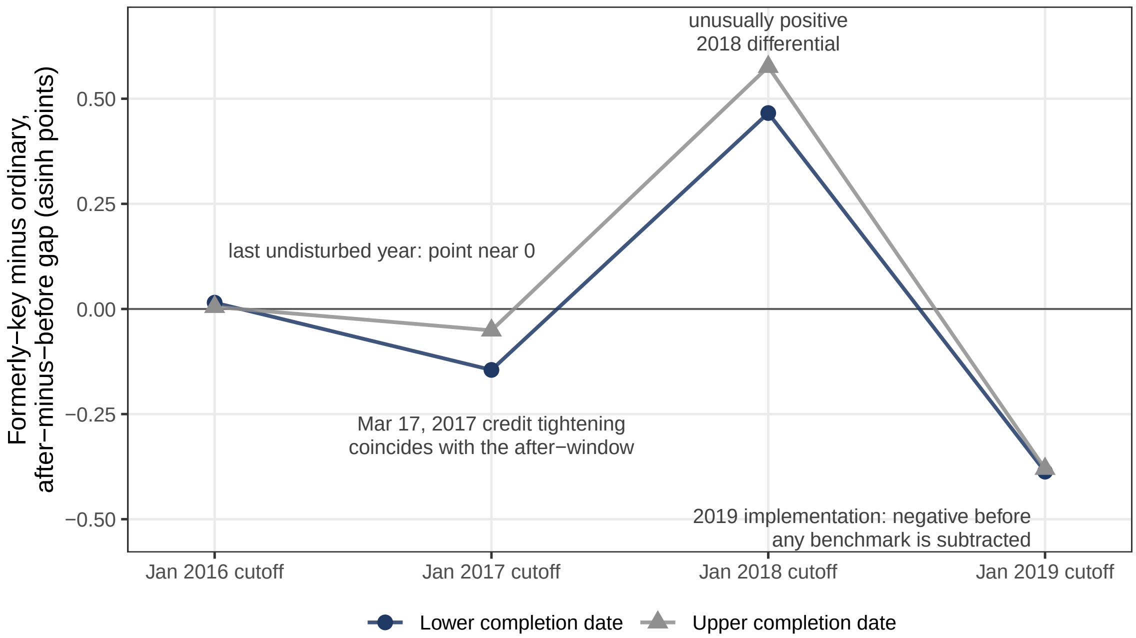


Figure 5: The seasonal benchmark, year by year

Notes: Each point is the formerly-key-minus-ordinary after-minus-before gap in completed transactions (asinh points) in a fixed-weight spatial frame chosen only from January 2015–August 2016 flows and coordinates. The 2018 point is an unusually positive differential; 2016 is the last undisturbed year, with a point estimate near zero. The 2019 gap is negative before any seasonal benchmark is subtracted.

### 5.3 Fact 2: The short-window divergence is not detectable cumulatively

A separately locked persistence exercise further narrows interpretation. The exposed-relative PPML ratio is 0.84 for calendar 2019 and 1.01–1.02 for calendar 2020. For cumulative 2019–2020 flow relative to 2016–2017, the ratio is 0.945 at both date endpoints, with asinh wild $p = 0.815$ and 0.835. The short-window differential is absent in this separately benchmarked cumulative comparison. This is consistent with delay or reallocation, but it is not a within-property convergence estimate; a similarly sized 2017 placebo also limits its use as causal validation.

Displacement into the control group is the sharpest spillover concern. The notice-verified controls sit close to the exposed communities—the median distance to the nearest exposed community is one kilometer and the maximum 2.5—so if the exposed shortfall reflected buyers substituting toward observationally similar ordinary-school housing, the rebound should be strongest in the nearest controls, and the triple difference would overstate the policy response. A pre-specified geographic diagnostic, reported in the online appendix, finds no detectable larger rebound nearby. Interacting the full experiment-side structure with a below-median-distance indicator in the control-only cell panel yields near-control effects of $-0.32$ and $-0.04$ (wild $p = 0.39$ and 0.92); the rank correlation between each control community's own rebound and its distance to the nearest exposed community is 0.07 and $-0.06$; and re-estimating the headline design against

Table 2: Long-horizon persistence and falsification

| Window and benchmark | PPML relative-flow ratio | | Asinh school-wild $p$-value | |
|---|---|---|---|---|
| | Lower | Upper | Lower | Upper |
| 16-week cutoff / identical 2018 calendar | 0.422 | 0.462 | 0.012 | 0.035 |
| Calendar 2019 / pooled 2015–2017 | 0.837 | 0.843 | 0.145 | 0.114 |
| Calendar 2020 / pooled 2015–2017 | 1.021 | 1.013 | 0.871 | 0.893 |
| Jan. 2019–Dec. 2020 / Jan. 2016–Dec. 2017 | 0.945 | 0.945 | 0.815 | 0.835 |
| 2017 placebo / pooled 2015–2016 | 0.881 | 0.887 | 0.121 | 0.193 |

Notes: Ratios below one denote lower formerly-key-relative completed flow. The first row reproduces the main sixteen-week design and exhausts all $2^{21}$ assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure. The remaining rows come from a separately locked persistence protocol estimated after the core cutoff analysis and retain 9,999 restricted-null draws clustered by historically assigned school. Annual models compare each indicated year with pooled pre-policy years; the two-year model compares cumulative calendar windows. The 2019 annual PPML cluster-$t$ $p$-values are 0.063 and 0.064; the table reports ratios for scale rather than PPML significance. The validation gate fails because the 2017 placebo is similar to the calendar-2019 result.

only far controls, or excluding the twenty controls that share a bizcircle with an exposed community, leaves PPML ratios of 0.41–0.47. The diagnostic does not reveal immediate diversion into the nearest controls. It cannot rule out diffuse or delayed reallocation, which remains one explanation for the cumulative two-year result.

### 5.4 Fact 3: Flow and time move more than selected prices

Table 3 places the headline completed-flow response beside conditional outcomes estimated on the same notice-verified frame, with the same design weights, fixed effects, and assigned-school clustering. The conditional specifications, predicted signs, and inference procedure were frozen on August 27, 2026, before inspection, on the original sixty-one-community frame; Panel B re-evaluates that unchanged protocol on the headline frame so both panels share one sample.

Under the designated benchmark, completed-flow ratios are 0.42–0.46. Among completed sales, marketing duration rises 0.70 and 0.75 log points at the two date endpoints (wild $p = 0.238$ and 0.043; Holm-corrected 0.714 and 0.130 across the three conditional outcomes). Transacted prices per square meter move −0.011 and −0.016 log points, and list-to-transacted-price ratios move −0.013 and −0.012; neither family is statistically distinguishable from zero. These are selected-outcome estimates, not marketwide price or duration responses.

A frozen tail diagnostic, reported in the online appendix, is directionally consistent with more unusually slow completed trades but is imprecise at both endpoints.

The configuration is consistent with the model's selection-and-delay pattern, but it does not identify a unique mechanism. Equation (13) shows why removing marginal matches need not move the mean price among surviving trades. The positive duration point estimates are directionally consistent with slower clearance. Because duration conditions on eventual completion and the flow window truncates later exits, the data do not support a quantitative mapping between duration and completed-flow ratios.

A group-specific demand, amenity, expectations, or platform-coverage shock could also reduce relative completed flow. The small conditional price estimates do not rule out those alternatives, especially with confidence intervals of roughly ±8 percent. They do establish a narrower point: conditional transaction prices need not move in proportion to completed flow. Consistent with that caution, the pre-announcement design-weighted hedonic price gap between exposed and control communities is small and imprecise—+0.024 (s.e. 0.103) over 2015 through April 2018 and +0.041 (s.e. 0.095) from 2017—but is descriptive rather than a causal measure of school value.

Table 3: Incidence around the cutoff: trade, time, and the price of what trades

| | Lower date | Upper date |
|---|---|---|
| *Panel A. Completed transaction flow (community cells, $n$ = 236)* | | |
| Asinh triple difference | −0.918 (0.335) | −0.762 (0.376) |
| exhaustive wild-school $p$ | 0.012 | 0.035 |
| PPML completed-flow ratio | 0.422 | 0.462 |
| *Panel B. Conditional on a completed sale (transactions, $n$ = 1,173)* | | |
| Log(1 + marketing duration) | +0.699 (0.422) | +0.745 (0.274) |
| wild-school $p$ [Holm] | 0.238 [0.714] | 0.043 [0.130] |
| Log transacted price per $m^2$ | −0.011 (0.043) | −0.016 (0.041) |
| wild-school $p$ [Holm] | 0.795 [0.856] | 0.686 [1.000] |
| Log list-price-to-transacted-price ratio | −0.013 (0.016) | −0.012 (0.018) |
| wild-school $p$ [Holm] | 0.428 [0.856] | 0.534 [1.000] |

Notes: All coefficients are exposed-relative after-minus-before effects net of the 2018 seasonal contrast, in the notice-verified preferred frame with headline design weights and assigned-school clustering. Panel A repeats the headline completed-flow design of Table 1. Panel B applies the conditional-outcome protocol frozen on August 27, 2026 to the same frame: community and experiment-side fixed effects with lower-order exposure interactions; the price and list-to-transacted regressions add log size, construction year, and floor-plan controls. Wild-school $p$-values use the restricted-null studentized procedure with 9,999 draws; brackets apply a Holm correction across the three Panel B outcomes. On the original 61-community protocol frame the estimates are +0.733 and +0.775 (duration), −0.011 and −0.015 (price), and −0.012 (ratio) at the two endpoints. Panel B conditions on sale, so these are selected outcomes: under the framework the transacted-price response is not signed, and its null is not evidence about capitalization.

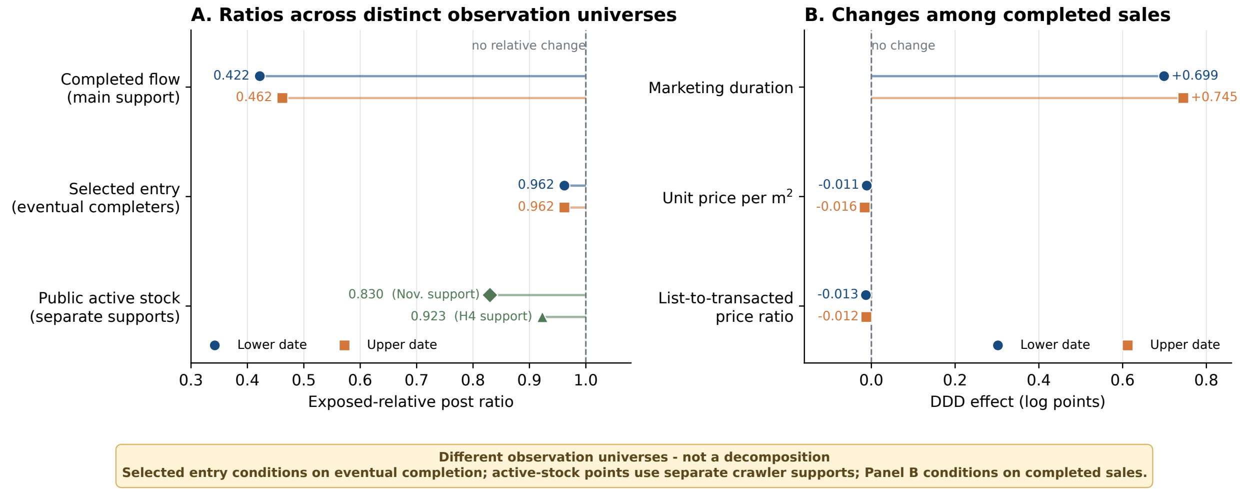


Figure 6: Where the response appears—and what each measure conditions on

Notes: Panel A reports exposed-relative ratios. Completed flow is the headline seasonal DDD; selected entry is defined only among listings known eventually to complete; the public-stock points come from separate active-listing cross-sections on different common supports. Panel B reports DDD coefficients among completed sales. Paired points are the two feasible completion-date endpoints. The panels organize point estimates; they do not place all outcomes on a common risk set or report a joint confidence region.

### 5.4.1 Transaction timing among eventual completers

The archive does not observe the full offered universe, but posting and completion dates permit exact stock-flow accounting within the subset of listings that eventually transact. This selected pipeline contains 1,364 posting events and 1,173 completed exits across 236 cells, and its accounting identity has zero error under both completion-date conventions. Entry has an exposed-relative PPML ratio of 0.962 under both dates, whereas exits reproduce the headline ratios of 0.422 and 0.462. The entry ratio is a point estimate, not an equivalence result; because the pipeline conditions on eventual completion, it does not measure full-market listing entry or a sale hazard.

Among 855 eventual sales with exact completion dates and unique outcome-blind links to the attention archive, marketing duration rises by 0.857 log-one-plus points (school-wild $p = 0.019$; CR2/Satterthwaite $p = 0.035$). Cumulative physical showings also rise, but daily follower, page-view, and physical-showing outcomes are imprecise. A scale-corrected physical-showing rate ratio is 0.728 ($p = 0.497$ against one), and its one-sided 95 percent lower bound of 0.328 fails the predeclared 75 percent noninferiority gate. The online appendix reports the complete frozen family and its original graphic. Within this selected pipeline, the duration estimate is consistent with later completion; it neither identifies a full-market completion hazard nor rules out listing-supply or reallocation explanations.

### 5.4.2 Public active stock and an assumption-indexed execution set

Public active-stock snapshots offer an independent comparison. On identical community supports, their exposed-relative point ratios lie closer to one than the corresponding completed-flow ratios, but the waves do not form a listing-transition panel. Let $R_F$ denote the exposed-relative ratio in completed flow, $R_E$ the ratio in period exposure to sale, and $R_h$ the ratio in execution rates. The identity $R_F = R_E R_h$ is accounting. The public crawls measure a point-stock contrast $\widetilde{R}_S$, not $R_E$. If $\kappa$ collects treatment-differential crawler coverage, the unobserved pre-period stock ratio, and point-stock-to-period-exposure alignment, then

$$R_h = \frac{R_F \kappa}{\widetilde{R}_S}, \qquad \mathcal{H}(K) = \left[ \frac{R_F \underline{\kappa}}{\widetilde{R}_S}, \frac{R_F \overline{\kappa}}{\widetilde{R}_S} \right]. \tag{4}$$

This is partial identification indexed by an analyst-specified comparability set $K$, not estimation of $\kappa$.

On the November-source support, the stock point is 0.830 while completed-flow ratios are 0.420 and 0.464. On the H4 support, the stock point is 0.923 while flow ratios are 0.368 and 0.427. Setting $\kappa = 1$ maps these points to execution ratios of approximately 0.40–0.56; a point fit with no execution difference requires $\kappa$ between 1.79 and 2.51. The online appendix reports the source table and full sensitivity curves. The exercise exposes the assumption required for an execution interpretation but does not separate buyer arrival, match acceptance, withdrawal, listing entry, or bargaining.

# 6 A Model of Trade with Nonportable Rights

The model formalizes the four-step spine previewed in the introduction. When the prior mapping is at least as valuable as the expected post-transfer bundle, a title reset raises the bilateral match threshold; the higher threshold removes marginal matches, changes which prices are observed, and slows finite-horizon clearance. The same primitives then compare where transition protection can attach. The reduced form does not estimate the model's primitives; it disciplines the ingredients and policy comparisons the model may contain. The framework is deliberately partial equilibrium and uses standard housing-search and bargaining logic (Wheaton, 1990; Genesove and Han, 2012; Head et al., 2014; Han and Strange, 2015). Seller reservation values, aggregate demand, seller entry, withdrawal, and congestion can produce similar flow-price-duration patterns (Stein, 1995; Genesove and Mayer, 2001; Anenberg and Ringo, 2024, 2026). What is setting-specific is the legal source of the transfer-contingent asymmetry.

## 6.1 Bilateral surplus when transfer resets eligibility

Consider a homeowner deciding whether to retain a dwelling and a prospective buyer considering the same dwelling. Their reservation value and willingness to pay are

$$R_s = \bar{R}_s + a_s q_0, \tag{5}$$

$$V_b = \bar{V}_b + a_b \mu_1, \tag{6}$$

where $q_0$ is the value of the prior single-school mapping conditional on enrollment eligibility, $\mu_1 = \mathbb{E}[Q_1]$ is the expected value of the post-transfer assignment bundle, and $a_s, a_b \geq 0$ are household-specific school-use values. Moving costs, housing-match quality, and outside options are contained in $\bar{R}_s$ and $\bar{V}_b$. Define entitlement-free match surplus $X = \bar{V}_b - \bar{R}_s$ and the match threshold under title-contingent grandfathering

$$\Delta^T = a_s q_0 - a_b \mu_1. \tag{7}$$

The threshold has a useful decomposition:

$$\Delta^T = \underbrace{(a_s - a_b) q_0}_{\text{ordinary heterogeneity in use values}} + \underbrace{a_b (q_0 - \mu_1)}_{\text{buyer-side title-reset component}}. \tag{8}$$

The first term can arise even when an amenity is fully transferable. The second arises because the assignment bundled with the property changes when title transfers. It is this second component that distinguishes the policy from an ordinary capitalization problem. A bilateral match has nonnegative surplus if and only if

$$X \geq \Delta^T. \tag{9}$$

The superscript $T$ denotes title-contingent grandfathering. The same physical house can therefore be mutually acceptable when the school right transfers and unacceptable when a sale resets eligibility. By contrast, an amenity change that shifts $R_s$ and $V_b$ by the same amount is capitalized into the transaction-price level while leaving match surplus, and hence equation (9), unchanged. That distinction separates the paper's transaction wedge from a standard capitalization exercise.

Conditional on predetermined property and market states, let $X$ have cumulative distribution $F$, density $f$, and survivor function $\bar{F} = 1 - F$.

**Proposition 1** (A higher threshold removes marginal matches)**.** *The probability that a potential match has nonnegative surplus is*

$$p(\Delta) = \mathbb{P}(X \geq \Delta) = \bar{F}(\Delta). \tag{10}$$

*At an interior point with $f(\Delta) > 0$, $p'(\Delta) = -f(\Delta) < 0$. An increase from $\Delta_0$ to $\Delta_1$ removes the set $\{X : \Delta_0 \leq X < \Delta_1\}$ from the positive-surplus region.*

*Proof.* Differentiate $1 - F(\Delta)$ with respect to $\Delta$. The discrete change is $F(\Delta_1) - F(\Delta_0)$. □

The model delivers a cross-sectional prediction only after an exposure ordering is supplied. If replacing a prior key-school mapping produces a larger expected loss than replacing an ordinary-school mapping, the post-transfer wedge rises more in the formerly key-school group. Completed activity in those communities can then fall behind ordinary-school communities. The empirical indicator supplies this ordering; it is not a household-specific estimate of $\Delta^T$.

### 6.2 Selection into completed prices

Suppose successful matches divide surplus through Nash bargaining and the seller receives share $\theta \in (0, 1)$. The transaction price is

$$P = R_s + \theta(V_b - R_s) = R_s + \theta(X - \Delta). \tag{11}$$

Prices are observed only when $X \geq \Delta$. Let $m(\Delta) = \mathbb{E}[X \mid X \geq \Delta]$, $e(\Delta) = m(\Delta) - \Delta$ denote mean excess surplus, and $h_X(\Delta) = f(\Delta)/\bar{F}(\Delta)$ denote the hazard of $X$.

**Proposition 2** (The trade response is signed; the selected-price response is not)**.** *Holding $R_s$ and $\theta$ fixed,*

$$\mathbb{E}[P \mid X \geq \Delta] = R_s + \theta e(\Delta), \tag{12}$$

$$\frac{\partial \mathbb{E}[P \mid X \geq \Delta]}{\partial \Delta} = \theta\{h_X(\Delta)e(\Delta) - 1\}. \tag{13}$$

*Thus a larger wedge reduces the mass of acceptable matches but can lower, leave unchanged, or raise the mean price among completed trades.*

*Proof.* Differentiating the conditional tail mean gives $m'(\Delta) = h_X(\Delta)[m(\Delta) - \Delta] = h_X(\Delta)e(\Delta)$. Because $e'(\Delta) = m'(\Delta) - 1$, substitution into equation (11) gives equation (13). □

For an exponential surplus distribution, mean excess surplus is constant and the selected mean price does not move locally even though acceptable matches disappear. Other distributions can generate either sign. A completed-sale hedonic regression therefore cannot recover the transaction wedge by itself, and controlling completed flow for a post-policy transaction price would condition on an equilibrium outcome.

### 6.3 Clearance and the horizon of the response

Suppose potential buyers arrive at Poisson rate $\lambda$ and draw independent match surpluses from $F$. Conditional on a property being continuously offered, the clearance rate is

$$h_C(\Delta) = \lambda \bar{F}(\Delta), \tag{14}$$

and the probability of completion by horizon $T$ is

$$H(T, \Delta) = 1 - \exp\{-\lambda \bar{F}(\Delta) T\}. \tag{15}$$

**Proposition 3** (A wedge delays trade without requiring permanent missing trade)**.** *For fixed $\lambda$ and $T > 0$, a larger $\Delta$ lowers both the clearance rate and finite-horizon completion probability:*

$$\frac{\partial H(T, \Delta)}{\partial \Delta} = -\lambda T f(\Delta) \exp\{-\lambda \bar{F}(\Delta) T\} < 0. \tag{16}$$

*Expected untruncated time to clearance is $1/h_C(\Delta)$ and rises with $\Delta$. If two groups have strictly positive clearance rates, their cumulative-completion ratio converges to one as $T \to \infty$.*

*Proof.* Differentiate equations (14) and (15). The limit follows because $H(T, \Delta) \to 1$ for every strictly positive $h_C(\Delta)$. □

This benchmark produces a sharp empirical implication: a large short-window divergence and a longer marketing clock can coexist with little cumulative deficit at a sufficiently long horizon. It does not assert that every property remains continuously listed or that all trades eventually occur. Withdrawals, seller entry, changing beliefs, and spatial reallocation can generate the same aggregate pattern, so the two-year result is a separately benchmarked qualitative consistency check, not an estimate or test of convergence in $H(T, \Delta)$.

To make that boundary explicit, let $L(\Delta)$ denote listing inflow and $W(\Delta)$ the withdrawal hazard. Active offered stock $S$ evolves as

$$\dot{S} = L(\Delta) - [h_C(\Delta) + W(\Delta)]S, \qquad S^*(\Delta) = \frac{L(\Delta)}{h_C(\Delta) + W(\Delta)}. \tag{17}$$

The sign of the stock response is unrestricted: a wedge can reduce seller entry and clearance simultaneously. A stock snapshot therefore cannot separate entry, acceptance, and withdrawal, while a completed-transaction archive contains exits but no full-market risk-set denominator.

### 6.4 How the facts discipline the model

Table 4 makes the fact-to-model mapping explicit. The model is asked to organize the joint configuration, not to convert each point estimate into a structural parameter.

Table 4: How the empirical facts discipline the model

| Empirical fact | Model object | Discipline and remaining margin |
|---|---|---|
| One-year completed-flow ratio 0.42–0.46 | Acceptance and clearance, $\bar{F}(\Delta)$ and $h_C(\Delta)$ | Requires a mechanism that removes or delays marginal matches; does not separate buyer arrival from seller entry |
| Conditional price and discount coefficients near zero | Selected mean in equation (13) | Rules out reading price as a sufficient statistic for the wedge; does not identify entitlement value |
| Positive selected-duration point estimates; eventual-completer entry ratio 0.96; active-stock ratios 0.83–0.92 | Clearance clock and stock law | Consistent with delayed execution; the three measures have different risk sets and do not identify a common listing hazard |
| Calendar-2020 ratio near one and cumulative two-year ratio 0.945 | Finite-horizon completion in equation (15) | Consistent with a short-run timing or reallocation margin; the aggregate comparisons neither track the same properties nor isolate convergence |

## 6.5 From the wedge to transition design

The model compares four transition rules. Under title-contingent grandfathering, the wedge is $\Delta^T$ in equation (7). If the old assignment follows the property after sale, if the incumbent household can carry it after moving, or if old and new owners face a uniform post-transition rule, respectively, the corresponding wedges are

$$\Delta^P = a_s q_0 - a_b q_0, \tag{18}$$

$$\Delta^H = -a_b \mu_1, \tag{19}$$

$$\Delta^U = (a_s - a_b)\mu_1. \tag{20}$$

Property portability raises what the buyer obtains from the dwelling. Household portability removes the incumbent's lost entitlement from the reservation value of selling. A uniform transition removes the title-vintage difference, but changes the incumbent's protected bundle even without a sale.

**Proposition 4** (Portability weakly expands the match set)**.** *Suppose $q_0 \geq \mu_1$ and $a_s, a_b \geq 0$. Holding school capacity, assignment probabilities, and outside housing options fixed,*

$$\Delta^T \geq \Delta^P, \qquad \Delta^T \geq \Delta^H, \qquad \Delta^T \geq \Delta^U. \tag{21}$$

*Hence each portable or uniform rule weakly raises the bilateral match probability relative to title-contingent grandfathering. The ordering among P, H, and U is generally ambiguous.*

*Proof.* The three differences are $\Delta^T - \Delta^P = a_b(q_0 - \mu_1)$, $\Delta^T - \Delta^H = a_s q_0$, and $\Delta^T - \Delta^U = a_s(q_0 - \mu_1)$, all weakly nonnegative. Apply Proposition 1. □

This is a partial-equilibrium match-rate ordering, not a welfare ranking. Property portability can preserve capitalization of a scarce school advantage in legacy homes. Household portability can create travel and destination-capacity costs. A uniform transition can violate the reliance protection that motivated grandfathering. Those general-equilibrium and distributional margins require preferences, realized assignments, capacity, and migration choices that the present data do not observe. The proposition nevertheless yields a concrete design principle: if policymakers wish to protect reliance, the protection need not expire solely because the household trades its home.

# 7 Policy Design: Where Should Protection Attach?

## 7.1 A design principle for grandfathering

The policy tension is not grandfathering per se. It is grandfathering by continued possession: the incumbent remains under the old rule while retaining the pre-cutoff title, and the protected rule expires when transfer generates a new title. A transition rule can protect reliance without making sale the forfeiture event.

Proposition 4 isolates a narrow transition-design margin. If policymakers want to honor reliance interests while preserving bilateral exchange, the legacy right need not expire solely because the asset is sold. Under $q_0 \geq \mu_1$ and holding household types, assignment probabilities, school capacity, and outside housing options fixed, property portability, household portability, and a uniform transition each weakly lower the match threshold relative to title-contingent grandfathering. Table 5 shows exactly which side of the match each rule changes.

Table 5: What should be grandfathered? Exchange margins and policy trade-offs

| Rule | What happens at sale | Exchange-margin result | Main unmeasured trade-off |
|---|---|---|---|
| Title-contingent | Incumbent loses $q_0$; buyer receives $\mu_1$ | Benchmark threshold $\Delta^T = a_s q_0 - a_b \mu_1$ | Sorting into legacy titles; capacity attached to old addresses |
| Property portability | Assignment $q_0$ follows the dwelling | Lowers threshold by $a_b(q_0 - \mu_1)$ | Perpetuates address-based advantage and capitalization |
| Household portability | Incumbent carries $q_0$; buyer receives $\mu_1$ | Lowers threshold by $a_s q_0$ | Travel, cross-neighborhood use, and destination capacity |
| Uniform transition | Incumbent and buyer both face $\mu_1$ | Lowers threshold by $a_s(q_0 - \mu_1)$ | Weakens reliance protection; changes congestion and incidence |

Notes: Threshold differences follow from Proposition 4 under $q_0 \geq \mu_1$ and $a_s, a_b \geq 0$. They order bilateral match probabilities in partial equilibrium, not welfare. Quantitative evaluation requires preferences, assignment probabilities, capacity, realized school outcomes, and equilibrium sorting.

Which portable rule is appropriate depends on what the transition is meant to protect. Property portability preserves the address-based entitlement and lets a buyer obtain it through the house price, but can perpetuate the spatial scarcity rent. Household portability protects the incumbent family while allowing the dwelling to enter the new regime, but can create travel and destination-capacity costs. A common date-based transition restores symmetry when equal treatment is valued more than legacy protection. A time-limited household entitlement is one implementation worth evaluating, not a rule selected by the model or the data.

## 7.2 Scale, mechanism, and scope

**An external transfer-tax yardstick.** A sale-contingent entitlement reset enters bilateral surplus like a heterogeneous transaction cost, although it is not a statutory tax, raises no public revenue, and varies with household demand for schools. Transfer-tax studies show that modest sale-contingent wedges can move transaction volume and timing sharply (Dachis et al., 2012; Best and Kleven, 2018). These estimates provide an external mechanism benchmark, not a conversion of the Haidian coefficient into an implicit tax rate. California's household-portability provision for Proposition 13, which reduced measured lock-in among newly eligible movers, illustrates a different attachment rule in practice (Ferreira, 2010).

**What the evidence identifies.** The title clause supplies a direct reason why transfer, rather than a common change in neighborhood quality, can alter match surplus. The joint pattern—a short-run completed-flow divergence, small conditional price and discount point estimates, positive selected-duration point estimates, and a separately benchmarked long-horizon comparison near one—is consistent with a title-reset mechanism. It is not point identification. The reduced-form design does not separately recover buyer arrival, seller participation, match acceptance, bargaining, listing supply, withdrawal, platform coverage, or spatial reallocation. Doing so would require a common listing panel with entry, viewing, offer, withdrawal, relisting, and terminal-sale states. The result also does not imply price rigidity: selection can move the mean price among completed trades in either direction. Welfare requires realized assignments, household preferences, migration choices, capacity, and equilibrium responses, as in structural residential-choice and school-admission work (Bayer et al., 2007; Cui et al., 2026).

External validity is local to nineteen communities attached to seven schools classified as key in one district. The estimate is a formerly-key-versus-ordinary differential response to replacing a prior single-school mapping, not the average effect of multi-school zoning on Haidian or Beijing. Its broader relevance lies in markets where ownership bundles a scarce public benefit and a known eligibility rule makes that benefit nontransferable.

# 8 Conclusion

Beijing's title-vintage rule protected incumbent school-assignment rights by making title transfer the event that extinguished them. Around implementation, formerly key-school communities recorded roughly six completed sales per community both before and after the cutoff, while notice-verified ordinary-school communities rose from roughly three to five. The designated one-year seasonal comparison implies completed-flow ratios of 0.42–0.46, but the unusually positive 2018 benchmark produces the largest magnitude; older and averaged benchmarks preserve the negative sign while shrinking the estimate.

The timing pattern is the central puzzle. There is no detectable broad pre-deadline acceleration, a sharp relative gap appears in the sixteen-week implementation comparison, and a separately benchmarked cumulative comparison yields a two-year ratio of 0.945 with no detectable differential. Prices and discounts among selected completed trades move little in point estimates while selected marketing-duration point estimates rise. A search-and-bargaining model organizes this configuration: when the prior mapping is more valuable than the expected post-transfer bundle, the title reset can remove matches near the surplus threshold; selection decouples the mean observed price from the mass of trade; and slower clearance can create a finite-horizon gap. The evidence is strongest as a short-run relative exchange response, not permanent lock-in, an observed convergence path, or a structural estimate of the entitlement's value.

The transition-design lesson is that grandfathering can be indexed to different objects. When a protected rule expires at sale, exchange itself becomes the costly state change. Under the stated partial-equilibrium assumptions, property portability, household portability, and a uniform transition each weakly expand the match set relative to title-contingent protection. A time-limited household entitlement is one concrete benchmark for evaluation when the objective is to protect an incumbent family rather than a legacy address; neither the model nor the data select it as optimal. Ranking transition rules requires assignment, capacity, preference, and migration data not observed here.

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# Online Appendix

Grandfathering at Sale: School Assignment Rights and Housing Exchange

Tai Lo Yeung

August 29, 2026

## A Model Details

Let the incumbent's reservation value and a prospective buyer's willingness to pay be

$$R_s = \bar{R}_s + a_s q_0, \tag{A1}$$

$$V_b = \bar{V}_b + a_b \mu_1, \tag{A2}$$

where $q_0$ is the deterministic entitlement preserved by retaining the old title, $\mu_1$ is the expected value of the post-transfer assignment bundle, and $a_s, a_b \geq 0$ are household-specific school-use values. Define entitlement-free match surplus $X = \bar{V}_b - \bar{R}_s$ and the transaction-contingent wedge

$$\Delta = a_s q_0 - a_b \mu_1. \tag{A3}$$

Trade occurs if and only if $X \geq \Delta$.

**Proposition A1** (A symmetric common shift leaves the model's trade set unchanged)**.** *If a policy changes buyer value and seller reservation value by the same amount, it changes the bargaining price but leaves the trade set unchanged.*

*Proof.* Adding the same scalar $d$ to $R_s$ and $V_b$ leaves $V_b - R_s$ and hence the positive-surplus indicator unchanged. □

Holding the valuation loadings fixed and treating $\Delta$ as a scalar threshold, suppose $X$ has cumulative distribution $F$ and density $f$. The probability that a potential match trades is $p(\Delta) = 1 - F(\Delta)$.

**Proposition A2** (A larger transaction-contingent wedge removes marginal matches)**.** *At any interior point with $f(\Delta) > 0$,*

$$\frac{\partial p(\Delta)}{\partial \Delta} = -f(\Delta) < 0. \tag{A4}$$

*An increase from $\Delta_0$ to $\Delta_1$ removes the set of matches $\{X : \Delta_0 \leq X < \Delta_1\}$.*

To characterize selection into observed prices, let the seller receive Nash weight $\theta$ and let $m(\Delta) = \mathbb{E}[X \mid X \geq \Delta]$. Conditional mean transaction price is

$$\mathbb{E}[P \mid X \geq \Delta] = R_s + \theta\{m(\Delta) - \Delta\}. \tag{A5}$$

Writing $e(\Delta) = m(\Delta) - \Delta$ and $h(\Delta) = f(\Delta)/[1 - F(\Delta)]$, and holding $R_s$ and $\theta$ fixed, gives

$$\frac{\partial \mathbb{E}[P \mid X \geq \Delta]}{\partial \Delta} = \theta\{h(\Delta)e(\Delta) - 1\}. \tag{A6}$$

The selected-price response is unsigned even though match probability falls. If potential buyers arrive at rate $\lambda$, a listing that remains offered clears at rate $\lambda[1 - F(\Delta)]$. A larger wedge lowers clearance and lengthens the expected marketing spell for fixed $\lambda$, but duration among eventual completers remains selected and active stock also depends on listing entry and withdrawal.

## B Measurement and Assignment Audit

The canonical transaction input is a processed proprietary listing-platform export containing 586,312 Beijing resale records posted from 2010 through January 2021. The policy analysis uses 2015–2020. All retained rows have positive transaction price, posting date, and marketing duration, so the archive contains completed transactions rather than an offered-listing denominator. The provider export does not document time-varying platform market share or off-platform transactions. The distributable package begins with the canonical analysis file and preserves every downstream policy transformation; reproducing the upstream provider-file cleaning step requires redistribution permission and a verified raw-to-canonical crosswalk.

When the exact completion date is missing because the source date is month-only, the paper retains the fixed interval from posting date plus reported duration minus one day to posting date plus duration. In the 414,979-record validation sample that observes both clocks, all but one reconstructed date lies within one day of the reported date. In the broader matched frame, 27 of 2,515 reconstructed dates (1.1 percent, or 0.36 percent of all 7,410 transactions) change cutoff-cell assignment across endpoints.

Table A1: Transaction archive and completion-date measurement

*Panel A. Archive-wide validation*

| Record universe | Count |
|---|---|
| Beijing completed-transaction archive | 586,312 |
| Transactions observing exact and reconstructed clocks | 414,979 |
| Reconstructed dates changing cutoff windows (broader frame) | 27 of 2,515 (1.1%) |

*Panel B. Retained-sample dates*

| | Preferred assignment frame | Exact-notice sensitivity |
|---|---|---|
| Completed transactions in four estimation windows | 1,173 | 1,029 |
| Resolved archive support in retained communities | 7,340 | 6,519 |
| Archive records with reported exact date | 4,863 | 4,303 |
| Archive records with reconstructed date | 2,477 | 2,216 |

Notes: The archive consists entirely of completed transactions. Reported exact dates are retained. Missing exact dates use the two fixed endpoints implied by posting date and marketing duration. The 414,979-record validation observes both clocks; all but one reconstructed date is within one day of the reported completion date. The broader-frame cutoff-window statistic has denominator 2,515 reconstructed dates (0.36 percent of all 7,410 records change windows).

Figure A1 reports an unsmoothed calendar-month diagnostic under both endpoints of the fixed completion-date interval. It shows neither a broad announcement-to-cutoff acceleration nor a literal one-month break at January 1; the relative ordinary-community rebound develops over the subsequent spring. The plot is deliberately not interpreted as an event study. Calendar-month assignment is much more sensitive to the one-day convention than the main sixteen-week cells: 232 reconstructed transactions move to an adjacent month between April 2018 and May 2019, whereas only 27 reconstructed transactions change membership in the main cutoff cells.

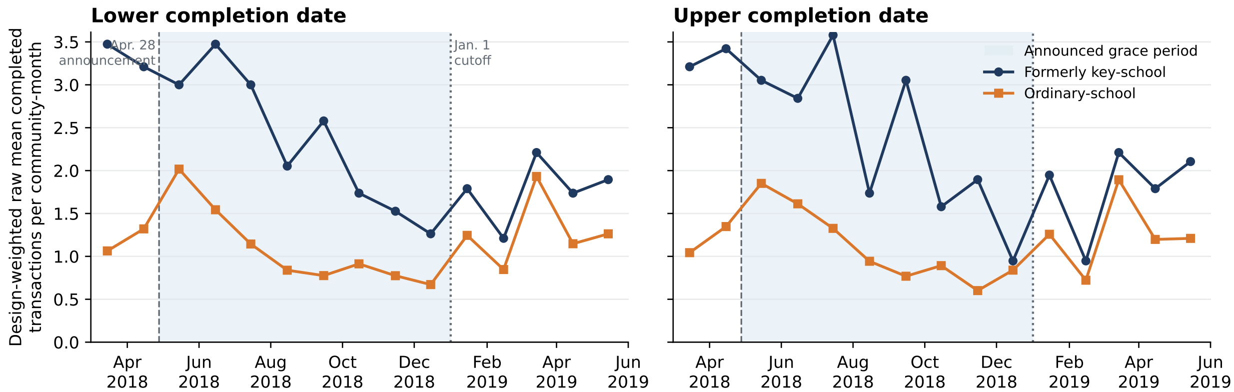


Figure A1: Raw monthly completed transactions under the two completion-date conventions

Notes: Each point is the inverse-reuse-weighted raw mean number of completed transactions per community-month in the preferred 59-community notice-verified frame. Zero-flow community-months are retained; no smoothing or significance filtering is applied. The two panels assign interval-dated transactions to the lower and upper endpoints of the fixed one-day completion interval. Raw group levels are not causal effects: both groups entered multi-school zoning, formerly key-school communities have higher baseline flow, and ordinary communities may receive reallocated transactions. April 2018 contains only three post-announcement days, and May 2019 lies beyond the formal sixteen-week post-cutoff window. Dates are platform-recorded completions rather than administrative title registrations.

The assignment frame uses visibly dated 2017 school-admission notices for every exposed community. The preferred comparison sample retains communities named literally in those notices or linked through an exact and unambiguous property or address alias. The exact-notice sensitivity retains only literal text matches. These restrictions preserve all nineteen exposed communities and do not rematch the design.

After freezing the design, I conducted an outcome-blind archival reconstruction of 2018 notice images to test whether the treated assignment was a transitory 2017 classification. Direct image review verifies the same assigned school in both years for all nineteen exposed communities and all seven exposed schools; Table A2 reports the school-level support. The 2018 images are preserved with page URLs and SHA-256 hashes, but are hosted by a public secondary archive rather than an authenticated district database. The main exposure therefore remains the frozen 2017 assignment, and the two-year check is used only as provenance and stability validation. Control-notice coverage is incomplete in 2018, so no control is added, deleted, or relabeled using that archive.

Table A2: Two-year historical-assignment validation for exposed communities

| Historically assigned school | Exposed communities | 2017 notice | Same assignment in 2018 |
|---|---|---|---|
| Qiyi Primary School | 3 | Verified | 3 of 3 |
| Renda Affiliated Experimental Primary | 1 | Verified | 1 of 1 |
| Shangdi Experimental Primary School | 3 | Verified | 3 of 3 |
| CNU Affiliated Yuxin School | 2 | Verified | 2 of 2 |
| Yangfangdian Central Primary School | 1 | Verified | 1 of 1 |
| Zhongguancun No. 3 Primary School | 1 | Verified | 1 of 1 |
| Renda Affiliated Primary School | 8 | Verified | 8 of 8 |
| Total | 19 | 19 of 19 | 19 of 19 |

Notes: The design is fixed from the 2017 notices. The 2018 column is a subsequent, outcome-blind stability audit based on direct review of dated notice images and does not rematch or select the estimation sample. Source pages, retained images, direct-read excerpts, and hashes are recorded in the replication archive. The images are hosted by a public secondary archive; the table therefore validates archival consistency, not district-database authentication.

Tables A3 and A4 provide stable row-level audit IDs and notice-image hash prefixes. The corresponding UTF-8 files in the replication folder preserve the original community and school names, matching phrase, public source URL, local image name, rationale, and full SHA-256 hash. Excluded candidates remain in the control audit rather than disappearing from the provenance trail.

Table A3: Exposed-community notice manifest

| Audit ID | School ID | Longitude | Latitude | Notice page | SHA-256 prefix |
|---|---|---|---|---|---|
| HT01 | 3 | 116.290 | 39.967 | 11330 | e3ce3d7c28 |
| HT02 | 3 | 116.288 | 39.969 | 11330 | e3ce3d7c28 |
| HT03 | 3 | 116.289 | 39.965 | 11330 | e3ce3d7c28 |
| HT04 | 3 | 116.295 | 39.969 | 11330 | e3ce3d7c28 |
| HT05 | 3 | 116.292 | 39.972 | 11330 | e3ce3d7c28 |
| HT06 | 3 | 116.285 | 39.969 | 11330 | e3ce3d7c28 |
| HT07 | 3 | 116.292 | 39.969 | 11330 | e3ce3d7c28 |
| HT08 | 3 | 116.285 | 39.967 | 11330 | e3ce3d7c28 |
| HT09 | 4 | 116.298 | 39.977 | 11268 | dadbcaffeb |
| HT10 | 30 | 116.358 | 40.071 | 12127 | 1927f52005 |
| HT11 | 30 | 116.352 | 40.062 | 12127 | 1927f52005 |
| HT12 | 46 | 116.338 | 39.914 | 11303 | 85eb12ddc1 |
| HT13 | 59 | 116.320 | 40.039 | 11342 | abf24ecb4b |
| HT14 | 59 | 116.323 | 40.041 | 11342 | abf24ecb4b |
| HT15 | 59 | 116.318 | 40.039 | 11342 | abf24ecb4b |
| HT16 | 61 | 116.305 | 39.895 | 11233 | 7e9d768cef |
| HT17 | 61 | 116.315 | 39.901 | 11233 | 7e9d768cef |
| HT18 | 61 | 116.313 | 39.902 | 11233 | 7e9d768cef |
| HT19 | 67 | 116.340 | 39.989 | 12175 | d27f2ba414 |

Notes: Stable audit IDs map to the UTF-8 community and school names, complete notice URLs, dates, image filenames, and full SHA-256 hashes in `Replication/frozen_matched_community_frame_assignment_audited.csv`. All nineteen records are visibly dated 2017 notice-image links. School IDs are stable internal identifiers, not rankings.

Table A4: Ordinary-community assignment audit manifest

| Audit ID | Evidence | Preferred | Literal | SHA-256 prefix |
|---|---|---|---|---|
| HC01 | E | Yes | Yes | a3e6ae419b |
| HC02 | U | No | No | 5acd502cb8 |
| HC03 | E | Yes | Yes | 5acd502cb8 |
| HC04 | E | Yes | Yes | 5acd502cb8 |
| HC05 | A | Yes | No | 5acd502cb8 |
| HC06 | E | Yes | Yes | 5acd502cb8 |
| HC07 | E | Yes | Yes | 5acd502cb8 |
| HC08 | E | Yes | Yes | 5acd502cb8 |
| HC09 | E | Yes | Yes | 5acd502cb8 |
| HC10 | E | Yes | Yes | b473d367d5 |
| HC11 | E | Yes | Yes | b473d367d5 |
| HC12 | E | Yes | Yes | b473d367d5 |
| HC13 | E | Yes | Yes | de7bd69caa |
| HC14 | E | Yes | Yes | 649ceaa638 |
| HC15 | E | Yes | Yes | 649ceaa638 |
| HC16 | E | Yes | Yes | 6cde492527 |
| HC17 | E | Yes | Yes | 6cde492527 |
| HC18 | E | Yes | Yes | 6cde492527 |
| HC19 | E | Yes | Yes | 6cde492527 |
| HC20 | E | Yes | Yes | 3016590a80 |
| HC21 | E | Yes | Yes | 3016590a80 |
| HC22 | E | Yes | Yes | 3016590a80 |
| HC23 | A | Yes | No | 3016590a80 |
| HC24 | E | Yes | Yes | 54886b022f |
| HC25 | E | Yes | Yes | 54886b022f |
| HC26 | E | Yes | Yes | 0376bdbe12 |
| HC27 | S | No | No | d7913db869 |
| HC28 | E | Yes | Yes | a8edee6900 |
| HC29 | E | Yes | Yes | a8edee6900 |
| HC30 | A | Yes | No | a8edee6900 |
| HC31 | E | Yes | Yes | 754729ffe5 |
| HC32 | A | Yes | No | 754729ffe5 |
| HC33 | A | Yes | No | 754729ffe5 |
| HC34 | E | Yes | Yes | 754729ffe5 |
| HC35 | A | Yes | No | 754729ffe5 |
| HC36 | A | Yes | No | 754729ffe5 |
| HC37 | E | Yes | Yes | d976c37825 |
| HC38 | A | Yes | No | 3c7e243264 |
| HC39 | E | Yes | Yes | 3c7e243264 |
| HC40 | E | Yes | Yes | 3c7e243264 |
| HC41 | A | Yes | No | 3c7e243264 |
| HC42 | E | Yes | Yes | 446569c893 |

Notes: E = literal notice text; A = exact address or unambiguous alias; S = secondary source only; U = unresolved. The preferred frame retains 31 E and 9 A records; the literal-notice sensitivity retains only E. The two excluded candidates remain visible. The full UTF-8 crosswalk—including community and school names, matching phrases, source URLs, rationales, local image names, and complete hashes—is `Replication/control_assignment_audit_2017.csv`.

The external formerly-key classification also has independent pre-announcement market content. A school component estimated from 2015–2016 transaction prices improves 2017 out-of-sample prediction, and the seven formerly-key schools have a higher component than the fourteen ordinary schools. Table A5 reports this treatment-proxy validation. A separately frozen continuous-dose policy specification fails its promotion gates and is not used to rescue the binary result.

Table A5: Pre-announcement market-value validation of the binary exposure proxy

| Validation quantity | Result |
|---|---|
| Historical schools with 2015–2016 support | 21 / 21 |
| Schools with at least five support transactions | 18 / 21 |
| 2017 transaction-level MSE reduction from school component | 4.67% |
| 2017 school-mean validation slope ($N$ = 20 schools) | 0.934 |
| One-sided validation-slope $p$-value | 0.0000062 |
| Formerly-key minus ordinary school-value dose | 0.851 SD |
| One-sided enumerated classification-label sensitivity $p$-value | 0.0462 |
| Continuous policy-dose promotion gate | Fails |

Notes: The school component is estimated only from 2015–2016 prices and validated on 2017 transactions before the policy outcome is opened. All twenty-one schools have training support; the validation-slope regression uses the twenty schools with 2017 transactions. The binary key-school classification is external to this exercise. The one-sided label sensitivity enumerates all 116,280 assignments of seven labels among twenty-one schools and is a treatment-proxy diagnostic, not randomization inference for the policy effect. The separately frozen continuous-dose policy interaction fails significance and placebo-dominance gates and is not used as an outcome specification.

Table A6 records the observation counts, communities, clusters, and identifying variation for the headline and supporting samples.

Table A6: Estimation samples and identifying support

| | Preferred assignment frame | Exact-notice sensitivity |
|---|---|---|
| Completed transactions in four estimation windows | 1,173 | 1,029 |
| Exposed communities | 19 | 19 |
| Exposed communities assigned to the same school in 2017 and 2018 | 19 of 19 | 19 of 19 |
| Comparison communities | 40 | 31 |
| Matched sets | 19 | 19 |
| Historically assigned-school clusters | 21 | 21 |
| Exposed-school clusters | 7 | 7 |
| Community–experiment–side cells | 236 | 200 |

Notes: The preferred frame retains comparison communities named in contemporaneous 2017 notices or linked by an exact, unambiguous address alias. The exact-notice sensitivity retains only literal text matches. A separate, outcome-blind review of dated 2018 notice images verifies the same assigned school for all nineteen exposed communities; it does not alter either sample. Both samples preserve all nineteen exposed communities and seven exposed schools. Each community contributes four observations: before and after January 1 in the 2018 seasonal comparison and the 2019 policy experiment.

# C Additional Identification and Robustness Evidence

Before reconstructing missing completion dates, I recorded two designs in timestamped internal protocols: an announcement-period acceleration and a sixteen-week cutoff comparison. These were not externally preregistered. The announcement estimate provides no evidence of the predicted acceleration. The cutoff estimate has the predicted sign at both endpoints and under both notice restrictions. It nevertheless failed the original promotion gate: only eleven of twenty endpoint-specific window and donut estimates have wild-cluster $p < 0.05$; the upper-endpoint pseudo-cutoff rank is 0.143 rather than below 0.10; and 27 of 2,515 reconstructed dates (1.07 percent) change cells across endpoints, exceeding a denominator-specific .5 percent ambiguity threshold even though they are only 0.36 percent of all matched transactions. An earlier exact-date screen had already shown the local pattern, so subsequent reconstruction and notice restriction validate a known secondary signal.

Two later additions follow the same discipline. First, a conditional-outcome protocol frozen on August 27, 2026, before inspection, fixed three outcomes on the cutoff cells (log one-plus marketing duration, hedonic log transacted unit price, and the log list-to-transacted ratio), their predicted signs, and restricted-null wild-school inference. The main-text incidence table re-evaluates those unchanged specifications on the headline notice-verified frame so that the flow and conditional panels share one sample; the original-frame estimates are nearly identical and are recorded in the table notes. Second, a geographic displacement diagnostic was pre-specified on August 29, 2026 with predictions and all design choices (nearest-exposed distance, the below-median split, and the bizcircle adjacency notion) fixed from geometry alone before any outcome was estimated; all four registered tests are reported in the main text regardless of outcome.

The raw 2019 cells are essential for interpreting that sign: completed counts are approximately flat in exposed communities and rebound sharply in ordinary-school communities. The cutoff coefficient is therefore a relative DDD shortfall, not an observed collapse in exposed transactions or a districtwide volume estimate. The absence of a broad Haidian grace-period response also differs from Xicheng, where Shao et al. (2023) report an announcement-period increase followed by an enforcement-period decline in transaction volume.

The main paper reports the preferred and literal-notice estimates with CR2 intervals and the exposed-school concentration. School sign counts are descriptive concentration diagnostics because controls are reused across matched sets.

The outcome-blind spatial match pairs all nineteen exposed communities to distinct notice-verified controls within 2.384 kilometers. School-equal post-cutoff effects are −0.718 and −0.728. Under a school-effect symmetry restriction, two-sided sign-flip sensitivities are 0.047 and 0.078; exact-notice controls produce 0.047 at both endpoints. These are not randomization-inference tests. Six of seven school effects are negative, every exposed-school omission preserves the negative sign, and symmetric Lunar-New-Year exclusions retain 78–96 percent of the corresponding magnitude. The 1.5 km rematch retains only twelve communities and six schools and is near zero; the 2.0 km match retains seventeen, and full support begins at 2.5 km. Baseline housing attributes are balanced, but the 2017 completed-flow standardized difference is about 0.40. The match is therefore corroboration of a local counterfactual, not a second independently decisive design.

The full-sample surface exercise permits the experiment-side movement to vary smoothly across Haidian. Linear surfaces leave the coefficients slightly larger in magnitude and below 0.05 under school-wild inference at both endpoints. Quadratic surfaces preserve the magnitudes but widen school-wild $p$-values to 0.105 and 0.138. Conley standard errors are reported only as residual-dependence diagnostics: residual Moran statistics are close to zero and the Conley standard errors fall as the distance cutoff expands, so they do not supply independent identifying variation.

A post-baseline balancing exercise uses the complete local donor universe rather than one control per exposed community. Nonnegative weights are chosen only from January 2015–August 2016 monthly completed flows and coordinates; a 2015 fit and January–August 2016 validation select the penalty before any

Table A7: Outcome-blind spatial-pair validation

| | Lower completion date | | | Upper completion date | | | Support |
|---|---|---|---|---|---|---|---|
| Specification | Estimate | Exact $p$ | CR1 $p$ | Estimate | Exact $p$ | CR1 $p$ | Pairs/schools |
| Exact/alias, 1.0 km | $-0.879$ | 0.188 | 0.173 | $-1.065$ | 0.125 | 0.089 | 7/5 |
| Exact/alias, 1.5 km | 0.040 | 0.938 | 0.926 | $-0.065$ | 0.906 | 0.883 | 12/6 |
| Exact/alias, 2.0 km | $-0.929^{**}$ | 0.047 | 0.066 | $-0.813^{**}$ | 0.047 | 0.077 | 17/7 |
| Exact/alias, 2.5 km | $-0.718^{**}$ | 0.047 | 0.108 | $-0.728^{*}$ | 0.078 | 0.116 | 19/7 |
| Exact/alias, 3.0 km (primary) | $-0.718^{**}$ | 0.047 | 0.108 | $-0.728^{*}$ | 0.078 | 0.116 | 19/7 |
| Exact-notice controls, 3.0 km | $-0.836^{**}$ | 0.047 | 0.077 | $-0.845^{**}$ | 0.047 | 0.078 | 19/7 |
| Nearest with replacement, 3.0 km | $-0.871^{**}$ | 0.016 | 0.045 | $-0.752^{*}$ | 0.094 | 0.121 | 19/7 |
| LNY blackout $\pm$14 days | $-0.564$ | 0.141 | 0.150 | $-0.565$ | 0.172 | 0.178 | 19/7 |
| LNY blackout $\pm$21 days | $-0.630$ | 0.203 | 0.184 | $-0.643$ | 0.219 | 0.225 | 19/7 |
| LNY blackout $\pm$28 days | $-0.692$ | 0.203 | 0.176 | $-0.693$ | 0.188 | 0.223 | 19/7 |
| Pre-policy placebo: 2018 versus 2017 | 0.459 | 0.281 | 0.323 | 0.470 | 0.344 | 0.365 | 19/7 |

Notes: Negative estimates use the paper's convention and denote an exposed-relative completed-flow shortfall. The match is maximum-cardinality, minimum-total-distance and uses only coordinates and the independent 2017 notice audit; except where stated, controls are distinct. The primary 3 km assignment was frozen before matched outcomes were opened. Estimates first average community-pair DDDs within exposed school and then average schools. Two-sided sensitivities enumerate all $2^G$ joint sign flips at the exposed-school level under a school-effect symmetry restriction; they are not randomization inference. Stars use these values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. CR1 $p$-values use $t(G-1)$. Each caliper is rematched, so changing support explains why the 1.5 km row is not comparable to a simple trimming exercise. LNY rows symmetrically remove the indicated days around the year-specific Lunar New Year. The placebo compares the January 2018 differential with January 2017.

Table A8: Balance in the frozen one-to-one spatial match

| Predetermined variable | Standardized mean difference |
|---|---|
| Longitude | $-0.244$ |
| Latitude | 0.037 |
| 2017 asinh completed flow, lower date | 0.404 |
| 2017 asinh completed flow, upper date | 0.399 |
| Median log unit price | 0.048 |
| Median floor area | 0.032 |
| Median construction year | 0.117 |

Notes: Differences are formerly-key minus ordinary, standardized by the pooled dispersion in the nineteen frozen pairs. Housing attributes are available for eighteen pairs. Matching uses coordinates and assignment provenance, not outcomes or these balance statistics. The approximately 0.40 pre-policy-flow difference motivates the later preperiod-balancing diagnostic and remains a limitation of the spatial comparison.

Table A9: Spatial robustness of the completed-flow triple difference

| Specification | Coefficient | School-wild $p$ | Conley $p$: 1/2/3 km |
|---|---|---|---|
| *Panel A. Lower completion date, full preferred sample* | | | |
| No spatial surface | $-0.918^{**}$ (0.335) | 0.012 | 0.015/0.007/0.003 |
| Linear geography × cells | $-0.972^{**}$ (0.372) | 0.021 | 0.021/0.009/0.004 |
| Quadratic geography × cells | $-0.905$ (0.432) | 0.105 | 0.026/0.005/0.001 |
| *Panel B. Upper completion date, full preferred sample* | | | |
| No spatial surface | $-0.762^{**}$ (0.376) | 0.035 | 0.057/0.039/0.030 |
| Linear geography × cells | $-0.839^{**}$ (0.395) | 0.049 | 0.053/0.030/0.021 |
| Quadratic geography × cells | $-0.766$ (0.448) | 0.138 | 0.070/0.031/0.018 |
| *Panel C. Previously locked outcome-blind 3 km spatial match* | | | |
| Lower completion date | $-0.718^{**}$ (0.380) | 0.047 | – |
| Upper completion date | $-0.728^{*}$ (0.397) | 0.078 | – |

Notes: The outcome is the inverse-hyperbolic-sine of completed transactions per community in a 16-week cell. Every coefficient uses the paper's After × 2019 × Formerly-key convention, so a negative value is an exposed-relative, seasonally netted shortfall. Panels A and B retain all 59 communities and the original inverse-reuse weights. All models contain community fixed effects, experiment-by-side fixed effects, and lower-order exposure interactions. Linear and quadratic models permit the stated coordinate surface to differ in every experiment-side cell. Parentheses contain assigned-school CR1 standard errors. The no-surface rows use exhaustive restricted-null wild inference over all $2^{21}$ assigned-school Rademacher sign vectors; the linear and quadratic rows retain their recorded 9,999-draw values. Conley columns report two-sided $t(N-K)$ $p$-values for Bartlett kernels at 1, 2, and 3 km. Panel C reproduces the locked school-equal spatial-pair benchmark; its final column reports the exhaustive seven-school sign-flip sensitivity under a symmetry restriction, not randomization inference. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

later outcome enters the final weights. The preferred weights retain all forty controls, have effective sample sizes of 29.6 and 29.8, and put at most 5.6 and 6.4 percent on one control. The coefficients remain negative in the paper's convention across the complete fixed-penalty family and every recomputed omission of any of the twenty-one represented schools.

The fully held-out 2018-versus-2017 placebo is nevertheless sizeable and opposite-signed, with wild $p = 0.068$ and 0.049. It is not a clean null. As the main text's benchmark discussion documents, the 2017–2018 sequence is consistent with a differential shock and recovery around Beijing's March 17, 2017 credit tightening, which coincides with the 2017 after-window and was plausibly most binding in the school-district segment; that timing does not establish the mechanism. The placebo is the oscillation viewed on its own, and the 2016 point estimate is near zero. Multiyear comparisons were added only after observing that result. Relative to the average 2017–2018 seasonal gap, the 2019 estimates remain $-0.547$ and $-0.642$; relative to 2017 alone, they are smaller and imprecise. Tables A10 and A11 preserve both the formal result and the adverse diagnostic.

For completeness, Table A12 collects the influence, spatial, and balancing exercises by design status. Figure A2 displays the complete bandwidth/donut family whose specifications were fixed before date reconstruction and all thirteen earlier pseudo-cutoffs. These exhibits are diagnostic rather than an independent research design.

The pre-specified nearby-control diagnostic and its far-control and shared-bizcircle sensitivities are reported in Table A14.

Table A10: Spatially local preperiod-balanced validation

| | Lower completion date | | | Upper completion date | | |
|---|---|---|---|---|---|---|
| Specification | Estimate | CR2 $p$ | Wild $p$ | Estimate | CR2 $p$ | Wild $p$ |
| CV-selected penalty | $-0.853^{**}$ | 0.034 | 0.015 | $-0.955^{***}$ | 0.028 | 0.009 |
| Fixed $\lambda = 0$ | $-0.873^{***}$ | 0.049 | 0.006 | $-0.876^{**}$ | 0.094 | 0.027 |
| Fixed $\lambda = 0.01$ | $-0.825^{**}$ | 0.045 | 0.013 | $-1.003^{***}$ | 0.023 | 0.004 |
| Fixed $\lambda = 0.1$ | $-0.853^{**}$ | 0.034 | 0.015 | $-0.955^{***}$ | 0.028 | 0.009 |
| Fixed $\lambda = 1$ | $-0.877^{**}$ | 0.021 | 0.011 | $-0.972^{***}$ | 0.022 | 0.008 |
| Fixed $\lambda = 10$ | $-0.881^{***}$ | 0.018 | 0.009 | $-0.978^{***}$ | 0.021 | 0.009 |
| *Earlier calendar experiments, CV-selected weights* | | | | | | |
| 2018 versus 2017 (fully held out) | $0.611^{*}$ | 0.080 | 0.068 | $0.627^{**}$ | 0.062 | 0.049 |
| 2017 versus 2016 (partly in sample) | $-0.160$ | 0.581 | 0.578 | $-0.056$ | 0.840 | 0.831 |

Notes: Negative estimates use the paper's convention and denote an exposed-relative completed-flow shortfall. Control weights are nonnegative, sum to one, and use all notice-verified controls within 3 km of at least one exposed community. The preferred penalty is chosen using January–August 2016 prediction after fitting on 2015; final weights use only January 2015–August 2016 monthly flows and coordinates. CR2 uses coefficient-specific Satterthwaite degrees of freedom. Wild $p$-values use 9,999 restricted-null draws over every represented exposed and control school; stars use those values. The design was formalized after an exploratory pilot. The fully held-out placebo is opposite-signed and is not a clean pretrend pass. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A11: Year-specific seasonal gaps and multiyear diagnostics

| | Lower completion date | | | Upper completion date | | |
|---|---|---|---|---|---|---|
| Year/contrast | Estimate | CR2 $p$ | Wild $p$ | Estimate | CR2 $p$ | Wild $p$ |
| *Panel A. Formerly-key-minus-ordinary after-minus-before gap by year* | | | | | | |
| 2016 | 0.015 | – | – | 0.005 | – | – |
| 2017 | -0.145 | – | – | -0.051 | – | – |
| 2018 | 0.466 | – | – | 0.576 | – | – |
| 2019 | -0.387 | – | – | -0.380 | – | – |
| *Panel B. 2019 relative to alternative seasonal benchmarks* | | | | | | |
| 2019 minus 2018 (formal) | $-0.853^{**}$ | 0.034 | 0.015 | $-0.955^{***}$ | 0.028 | 0.009 |
| 2019 minus 2017 | $-0.241$ | 0.345 | 0.325 | $-0.329$ | 0.229 | 0.194 |
| 2019 minus average of 2017–2018 | $-0.547^{**}$ | 0.059 | 0.036 | $-0.642^{**}$ | 0.046 | 0.017 |
| 2019 minus average of 2016–2018 | $-0.499^{**}$ | 0.085 | 0.049 | $-0.557^{**}$ | 0.068 | 0.034 |

Notes: Negative values use the paper's after-minus-before convention. Panel A reports the weighted formerly-key-minus-ordinary seasonal gap by year. Panel B reports 2019 relative to alternative benchmarks. Only 2019 minus 2018 belongs to the formal protocol. The remaining rows were added after observing the sizeable opposite-signed held-out placebo and must be read as post-protocol diagnostics. Weights remain fixed at the CV-selected preperiod values. Stars use all-school wild-cluster $p$-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A12: Robustness and influence diagnostics

| Diagnostic | Lower date | Upper date |
|---|---|---|
| Preferred notice-verified effect | $-0.918^{**}$ [0.012] | $-0.762^{**}$ [0.035] |
| Exact-notice-only effect | $-0.870^{**}$ [0.022] | $-0.715^{*}$ [0.056] |
| Unweighted asinh effect | $-0.753^{**}$ [0.027] | $-0.715^{**}$ [0.048] |
| Leave-one-community/school range | [-1.146, -0.750] | [-0.994, -0.595] |
| Negative matched-set effects | 16/19 | 16/19 |
| Negative exposed-school effects | 7/7 | 6/7 |
| Recorded window/donut range | [-0.987, -0.274] | [-0.924, -0.029] |
| Negative window/donut effects | 10/10 | 10/10 |
| Absolute rank of real cutoff among 14 dates | 1 of 14 | 2 of 14 |
| Full sample: linear spatial surface | $-0.972^{**}$ [0.021] | $-0.839^{**}$ [0.049] |
| Full sample: quadratic spatial surface | $-0.905$ [0.105] | $-0.766$ [0.138] |
| Spatial pairs: 3 km, exact/alias | $-0.718^{**}$ [0.047] | $-0.728^{*}$ [0.078] |
| Spatial pairs: 3 km, exact notice | $-0.836^{**}$ [0.047] | $-0.845^{**}$ [0.047] |
| Road-network pairs: 4 km, exact/alias | $-1.167^{**}$ [0.047] | $-1.177^{*}$ [0.078] |
| Spatial pairs: pre-policy placebo | 0.459 [0.281] | 0.470 [0.344] |
| Local preperiod-balanced controls | $-0.853^{**}$ [0.015] | $-0.955^{***}$ [0.009] |
| Balanced held-out placebo: 2018 vs. 2017 | $0.611^{*}$ [0.068] | $0.627^{**}$ [0.049] |
| Balanced 2019 vs. 2017–18 mean (post-protocol) | $-0.547^{**}$ [0.036] | $-0.642^{**}$ [0.017] |

Notes: Effects use the main paper's after-minus-before sign convention; negative values denote a lower formerly-key-minus-ordinary after-before contrast in 2019 than in the comparison experiment. They do not require formerly key-school counts to fall in levels. The preferred and exact-notice rows use exhaustive restricted-null wild inference over all $2^{21}$ assigned-school Rademacher sign vectors. Spatial-surface and balanced rows retain their recorded 9,999-draw assigned-school wild $p$-values; the unweighted row uses its assigned-school CR1 $p$-value. All original-design diagnostics preserve the original matching when units are omitted. The window/donut family contains seven symmetric windows and three 16-week donut exclusions for each timing endpoint. Rank 1 denotes the largest absolute coefficient among the real January 2019 cutoff and thirteen earlier pseudo-cutoffs. Linear and quadratic surfaces interact centered geography with every experiment-side cell. The separately frozen one-to-one spatial-pair rows use an exhaustive two-sided exposed-school sign-flip sensitivity under a symmetry restriction; this is not randomization inference. The road-network row rematches on shortest-path distance in the frozen 2018 OpenStreetMap graph; four kilometers is the narrowest recorded network caliper retaining all nineteen exposed communities. Balanced-control weights are selected only with January 2015–August 2016 flows and coordinates; their design was formalized after an exploratory pilot. The 2019-versus-2017–18 row was added after the opposite-signed held-out placebo was observed. Stars follow $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

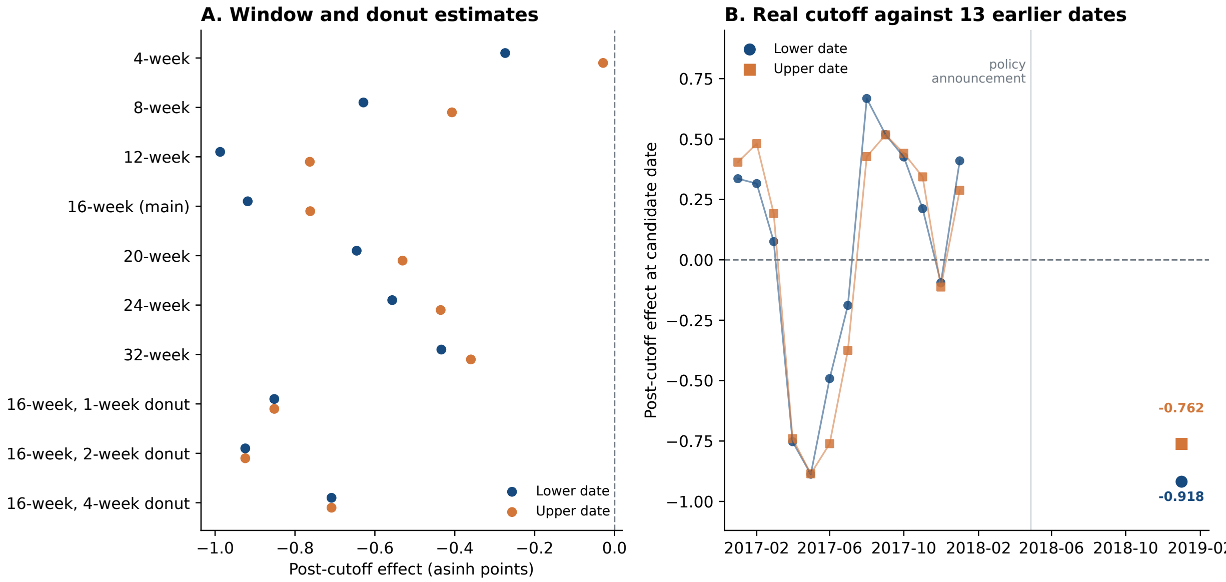


Figure A2: Specification and pseudo-cutoff diagnostics

Notes: Panel A displays every coefficient from the bandwidth/donut specification family fixed before date reconstruction. Panel B displays the real January 2019 coefficient and thirteen earlier pseudo-cutoffs. Coefficients are shown without significance filtering; preferred few-cluster inference appears in the main paper.

Table A13: Long-horizon persistence and falsification

| | PPML relative-flow ratio | | Asinh school-wild $p$-value | |
|---|---|---|---|---|
| Window and benchmark | Lower | Upper | Lower | Upper |
| 16-week cutoff / identical 2018 calendar | 0.422 | 0.462 | 0.012 | 0.035 |
| Calendar 2019 / pooled 2015–2017 | 0.837 | 0.843 | 0.145 | 0.114 |
| Calendar 2020 / pooled 2015–2017 | 1.021 | 1.013 | 0.871 | 0.893 |
| Jan. 2019–Dec. 2020 / Jan. 2016–Dec. 2017 | 0.945 | 0.945 | 0.815 | 0.835 |
| 2017 placebo / pooled 2015–2016 | 0.881 | 0.887 | 0.121 | 0.193 |

Notes: Ratios below one denote lower formerly-key-relative completed flow. The first row reproduces the main sixteen-week design and exhausts all $2^{21}$ assigned-school Rademacher sign vectors under the restricted-null, fully studentized wild procedure. The remaining rows come from a separately locked persistence protocol estimated after the core cutoff analysis and retain 9,999 restricted-null draws clustered by historically assigned school. Annual models compare each indicated year with pooled pre-policy years; the two-year model compares cumulative calendar windows. The 2019 annual PPML cluster-$t$ $p$-values are 0.063 and 0.064; the table reports ratios for scale rather than PPML significance. The validation gate fails because the 2017 placebo is similar to the calendar-2019 result.

Table A14: Geographic displacement diagnostics

| | Lower date | Upper date |
|---|---|---|
| Near-control extra rebound (controls only) | −0.322 (0.426) | −0.040 (0.450) |
| wild-school $p$ | 0.393 | 0.918 |
| Spearman rank correlation, control rebound vs. distance | +0.068 | −0.062 |
| Headline DDD, far controls only | −1.003 (0.414) | −0.773 (0.465) |
| wild-school $p$; PPML ratio | 0.021; 0.406 | 0.092; 0.467 |
| Headline DDD, excluding same-bizcircle controls | −0.930 (0.441) | −0.703 (0.483) |
| wild-school $p$; PPML ratio | 0.047; 0.417 | 0.155; 0.473 |

Notes: Pre-specified diagnostics for displacement of exposed-community transactions into nearby ordinary-school controls. Distance is from each control community to its nearest exposed community; the forty controls sit close to exposed communities (median nearest distance 1.00 km, maximum 2.54 km), so a substitution response should be visible in the near half. "Near" is below-median distance (20 of 40); the first row interacts the near indicator with the full experiment-side structure in the control-only cell panel, so a positive value would indicate rebound concentrated beside exposed communities. The middle row reports the rank correlation between each control community's own difference-in-differences and its distance. The lower rows re-estimate the headline design keeping only far controls, and excluding the 20 controls that share a bizcircle with an exposed community. Wild-school $p$-values use the restricted-null studentized procedure with 9,999 draws.

The matched pairs also permit a descriptive local decomposition. The formerly-key member's share of pair flow shifts by −10.4 and −11.5 percentage points, while combined pair flow shifts by −0.140 and −0.214 asinh points. Neither component is separately precise under the exhaustive seven-school sign-flip sensitivity, which requires school-effect symmetry and is not randomization inference. Because the nineteen pairs do not form a closed market, this diagnostic cannot distinguish local diversion, flows to unmatched locations, and aggregate suppression.

Table A15: Local spatial-pair total and share diagnostics

| | Lower completion date | | Upper completion date | |
|---|---|---|---|---|
| Outcome | Coefficient | Exact $p$ | Coefficient | Exact $p$ |
| Asinh pair-total transactions | −0.140 (0.321) | 0.688 | −0.214 (0.343) | 0.547 |
| Formerly-key transaction share (p.p.) | −10.372 (6.915) | 0.172 | −11.532 (6.776) | 0.125 |

Notes: The sample is the 19 distinct outcome-blind 3 km pairs. Each coefficient is the 2019 after-minus-before change net of the identical 2018 change, averaged first within treated school and then equally across the seven treated schools. Parentheses report CR1 treated-school standard errors; $p$-values and stars use the exhaustive two-sided $2^7$ school sign-flip sensitivity under school-effect symmetry. This is not randomization inference. A negative combined-total coefficient is a local pair-level shortfall. A negative share coefficient is a movement away from the formerly key member of the pair. The pairs do not form a closed market, so neither outcome identifies districtwide trade suppression. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

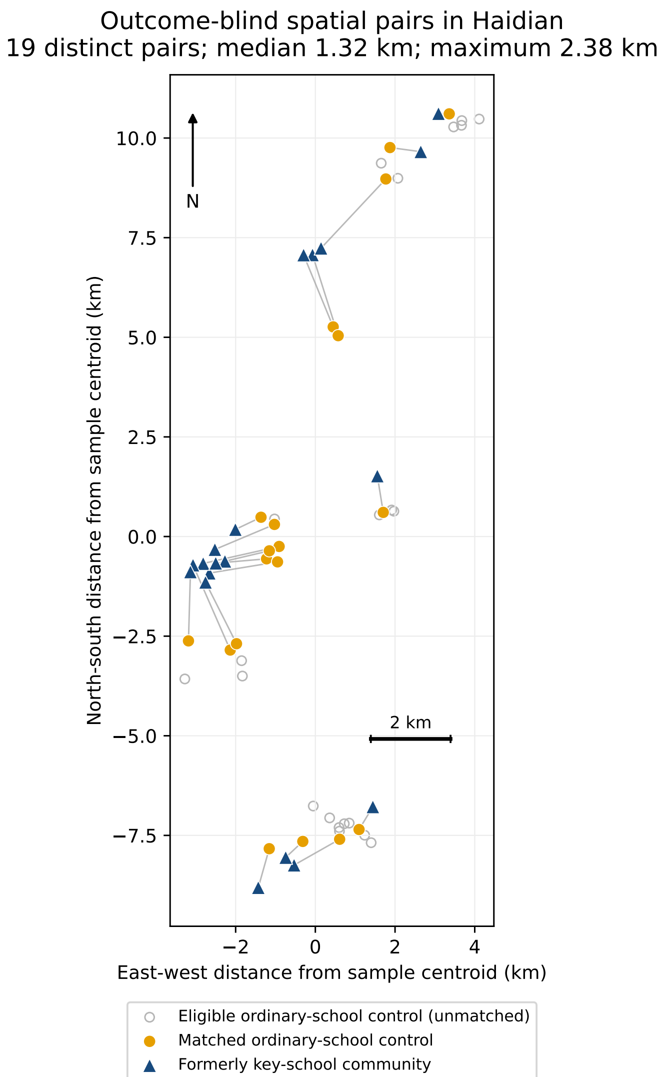


Figure A3: Outcome-blind one-to-one spatial pairs

Notes: Lines connect matched communities; they are not school-catchment boundaries. The global assignment maximizes retained exposed communities and then minimizes total geographic distance, without replacement, using only coordinates and the independent 2017 assignment audit. Coordinates share the same source coordinate system, so pair distances and relative positions are internally comparable.

Table A16: Analysis hierarchy and falsification checks

| Check | Lower timing endpoint | Upper timing endpoint | Interpretation |
|---|---|---|---|
| Notice-verified cutoff effect | $-0.918^{**}$ (wild p=0.012) | $-0.762^{**}$ (wild p=0.035) | Main completed-flow estimate |
| Exact-notice controls only | $-0.870^{**}$ (wild p=0.022) | $-0.715^{*}$ (wild p=0.056) | Assignment sensitivity |
| Monthly pre-period slope (61-community frame) | cluster-t p=0.029; wild p=0.121 | cluster-t p=0.059; wild p=0.190 | Inference differs with small-cluster correction |
| Real-cutoff rank among earlier dates | rank p=0.071 | rank p=0.143 | Only the lower endpoint falls below the 10 percent rank threshold |
| Announcement-to-cutoff acceleration | -0.380 (wild p=0.308) | -0.336 (wild p=0.392) | Estimate recorded before date repair |

Notes: Cutoff effects use the main-table sign, so negative values denote a lower formerly-key-minus-ordinary after-before contrast in 2019 than in the identical 2018 calendar experiment; they do not require formerly key-school counts to decline in levels. Stars use restricted-null wild-cluster inference: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The first two rows exhaust all $2^{21}$ assigned-school Rademacher sign vectors; remaining wild values retain their recorded Monte Carlo procedures. Lower and upper are the two endpoints of the fixed one-day reconstruction interval. Headline and exact-notice rows use the 59- and 50-community frames; rows explicitly labeled 61-community use the original broader frame. In the 414,979-transaction validation sample, all but one reconstructed date is within one day of the reported date. In the broader frame, 27 of 2,515 reconstructed dates (1.1 percent, or 0.36 percent of all 7,410 records) change cutoff-window membership across endpoints. All 20 endpoint-specific bandwidth/donut specifications were fixed before reconstruction and later produced the predicted sign, but only 11 have wild-cluster $p < 0.05$. The cutoff was a secondary estimand in timestamped internal protocols, not an external preregistration.

# D Completed-Pipeline and Public-Stock Accounting

Let $I_i = 1$ denote that listing $i$ eventually appears as a completed transaction. Define selected end-of-day active stock, entry, and exit as

$$S^C_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{a_i \le t < d_i\}, \tag{A7}$$

$$L^C_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{a_i = t\}, \tag{A8}$$

$$X_{ct} = \sum_{i:c_i=c} I_i \mathbf{1}\{d_i = t\}. \tag{A9}$$

They obey $S^C_{ct} = S^C_{c,t-1} + L^C_{ct} - X_{ct}$. The implemented identity has zero error in every community-period cell under both date conventions. This stock conditions on eventual completion and is not the full offered inventory.

Table A17 reports the full outcome family fixed before the attention coefficients were inspected, together with the subsequent scale-corrected physical-showing-rate audit. The duration component passes its predeclared sign and inference criterion, but the joint mechanism-promotion gate fails because corrected daily-rate noninferiority does not. Cumulative showings are consequently treated as exposure-loaded description rather than evidence of intact buyer arrival.

## D.1 A frozen selected-duration tail diagnostic

The mean selected-duration result motivates one distributional diagnostic. Before estimating its policy coefficient, I fixed the long-duration threshold at 85 days, the nearest-rank 90th percentile among preferred-

Table A17: Complete selected-completer outcome family

*Panel A. Entry and exit in the completed-listing pipeline*

| | Lower date | Upper date |
|---|---|---|
| Selected-entry PPML ratio | 0.962 | 0.962 |
| Completed-exit PPML ratio | 0.422 | 0.462 |
| Posting events / completed exits | 1,364 / 1,173 | |
| Community–experiment–side cells | 236 | |

*Panel B. Frozen exact-linked attention and duration outcomes*

| Outcome, log-one-plus scale | DDD | CR2 $p$ | Wild $p$ | $N$ / clusters |
|---|---|---|---|---|
| Physical showings per day | −0.069 | 0.432 | 0.402 | 855 / 19 |
| Followers per day | 0.064 | 0.584 | 0.551 | 855 / 19 |
| Page views per day | 0.148 | 0.739 | 0.716 | 755 / 17 |
| Cumulative physical showings | 0.854 | 0.064 | 0.040 | 855 / 19 |
| Marketing duration | 0.857 | 0.035 | 0.019 | 855 / 19 |

*Panel C. Scale-corrected physical-showing rate audit*

| | |
|---|---|
| Selected-completer PPML rate ratio | 0.728 |
| Two-sided school-clustered $p$-value against one | 0.497 |
| One-sided 95 percent lower rate-ratio bound | 0.328 |
| Wild one-sided $p$-value for $H_0$ : RR $\leq 0.75$ | 0.533 |
| Predeclared 75 percent noninferiority gate | Fails |

Notes: Every panel conditions on eventual completion and therefore does not measure the offered-listing universe. Panel B reports the complete frozen outcome family; coefficients are exposed-relative post effects after subtracting the identical 2018 calendar contrast. Wild tests use 9,999 restricted-null draws. The 855-record support contains 18 exposed and 38 control communities assigned to 6 exposed and 13 control schools; the page-view outcome has narrower support. Cumulative showings can rise mechanically with duration. Panel C uses a count model with listing days as exposure and supersedes an invalid noninferiority interpretation of the transformed showings-per-day outcome.

frame completed transactions with finite, nonnegative durations whose lower feasible completion date fell in 2015–2016; fixed the same weighted DDD, completion-date endpoints, school clustering, wild and CR2 inference; and required the predicted sign to survive every treated-school omission at both endpoints. Table A18 reports the complete prespecified estimate and inference family. The after-effect is positive at both endpoints and under all fourteen school-by-endpoint omissions, but the confidence intervals are wide. This diagnostic therefore locates the selected response in the right tail without establishing a marketwide duration distribution or a precise tail effect.

Table A18: Selected completed trades shift toward the long-duration tail

| | Lower completion date | Upper completion date |
|---|---|---|
| After-minus-before DDD (percentage points) | 10.2 | 18.1 |
| CR1S standard error (percentage points) | 12.9 | 9.7 |
| CR1S $p$-value | 0.437 | 0.077 |
| CR2 standard error (percentage points) | 14.7 | 10.6 |
| CR2 95% interval (percentage points) | [−27.0, 47.4] | [−8.7, 44.9] |
| School-wild $p$-value | 0.572 | 0.136 |
| CR2/Satterthwaite $p$-value | 0.516 | 0.145 |
| Treated-school leave-out range (percentage points) | [8.6, 12.3] | [15.8, 22.6] |
| Observations | 1,173 | 1,173 |
| Assigned-school clusters | 20 | 20 |

Notes: Before estimating the policy coefficient, the long-duration threshold was frozen at 85 days, the nearest-rank 90th percentile among 2,171 preferred-frame completed transactions with finite, nonnegative durations whose lower feasible completion date fell in 2015–2016. The outcome is one when recorded marketing duration is at least 85 days. Entries report the after-minus-before form of the weighted seasonal DDD; positive values indicate a larger exposed-relative long-tail share after implementation. CR1S denotes the conventional finite-cluster-corrected school-cluster covariance. The school-wild procedure uses 9,999 restricted-null draws. CR2 uses an identity working model with Satterthwaite degrees of freedom of 5.31 and 5.28. All seven treated-school omissions preserve the positive after-effect at both date endpoints. The sample conditions on completion and does not estimate the duration distribution of all offered listings.

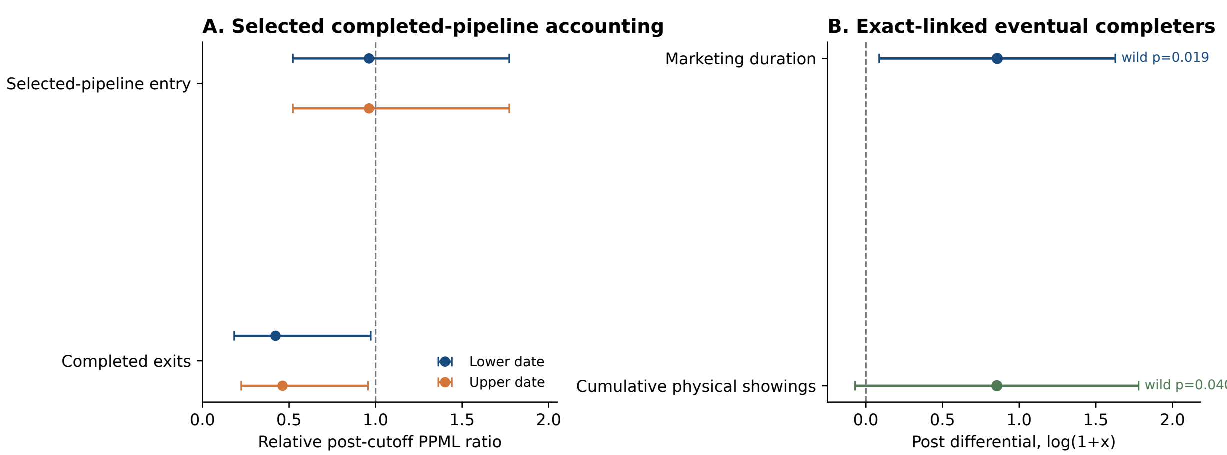


Figure A4: Timing and attention inside the eventual-completer pipeline

Notes: Panel A reports exposed-relative PPML ratios for entry into, and exit from, the set of listings known eventually to complete. Panel B reports real-versus-identical-calendar DDD estimates for the unique exact-linked eventual-completer sample. Intervals use the displayed equation-specific inference. Neither panel describes the full offered-listing universe.

The offered-stock comparison uses three independent public cross-sections: a November 28–29, 2017 crawl with exact listing IDs and a source-native community index (Huang, 2017); the H4M real-estate cross-section, whose contents are consistent with a late-March 2018 active-listing wave (Zhao et al., 2022); and a broad February 27, 2019 community-by-community crawl with 41,290 active rows across 4,131 named communities. Source-code review confirms iteration over every detected community page and every detected listing page, and the wave has positive rows for 54 of the 59 frozen communities. It nevertheless omits listing and dwelling IDs, row timestamps, and a request-success ledger, so it is not a certified universe. The union of the three waves is not a listing panel and does not reveal individual entry, withdrawal, relisting, or terminal states.

For the full offered universe, let $S^O_{ct}$ be end-of-period active stock, $E^O_{ct}$ the period's pre-exit exposure to sale, $L^O_{ct}$ new entries, $W^O_{ct}$ withdrawals, and $h_{ct}$ the execution rate. The accounting system is

$$E^O_{ct} = S^O_{c,t-1} + L^O_{ct}, \tag{A10}$$

$$X_{ct} = E^O_{ct} h_{ct}, \tag{A11}$$

$$S^O_{ct} = E^O_{ct} - X_{ct} - W^O_{ct}. \tag{A12}$$

For aligned exposed-comparison period ratios, this implies $R_F = R_E R_h$, where $R_F$ is relative completed flow, $R_E$ relative period exposure, and $R_h$ the relative execution rate.

The public snapshots measure a point-stock contrast $\widetilde{R}_S$, not period exposure $R_E$. Let $\kappa$ collect treatment-differential crawler coverage, the unobserved cross-year pre-period stock ratio, and point-snapshot-to-period-exposure alignment. Defining $R_E = \widetilde{R}_S/\kappa$ gives

$$R_h = \frac{R_F \kappa}{\widetilde{R}_S}. \tag{A13}$$

**Proposition A3** (Comparability-indexed execution set)**.** *If the combined comparability multiplier belongs to $K = [\underline{\kappa}, \overline{\kappa}]$, then, conditional on point values $(R_F, \widetilde{R}_S)$,*

$$\mathcal{H}(K) = \left[\frac{R_F \underline{\kappa}}{\widetilde{R}_S}, \frac{R_F \overline{\kappa}}{\widetilde{R}_S}\right]. \tag{A14}$$

*No execution change is compatible with the points if and only if $\widetilde{R}_S/R_F \in K$.*

On the November-source support, the exposed-relative stock point is 0.830 and the exposed-relative completed-flow ratios are 0.420 and 0.464. On the H4 support, the corresponding stock point is 0.923 and the completed-flow ratios are 0.368 and 0.427. Equality inference is correction-sensitive and neither source uniformly rejects a supply-only explanation under every finite-sample correction.

At $\kappa = 1$, the points imply execution-rate ratios of approximately 0.40–0.56. A supply-only point fit requires $\kappa$ between 1.79 and 2.51. The comparison makes the required stock adjustment transparent but does not separately identify buyer arrival, match efficiency, acceptance, withdrawal, or bargaining.

Table A19: Completed flow and public active stock on identical supports

| Common support | Object | Relative ratio | 95% interval | Communities |
|---|---|---|---|---|
| Nov. 2017–Feb. 2019 common support | Completed flow (lower date) | 0.420 | [0.167, 1.057] | 54 |
| Nov. 2017–Feb. 2019 common support | Completed flow (upper date) | 0.464 | [0.209, 1.033] | 54 |
| Nov. 2017–Feb. 2019 common support | Public active-stock contrast | 0.830 | [0.477, 1.447] | 54 |
| Late-Mar. 2018–Feb. 2019 H4 support | Completed flow (lower date) | 0.368 | [0.117, 1.154] | 37 |
| Late-Mar. 2018–Feb. 2019 H4 support | Completed flow (upper date) | 0.427 | [0.160, 1.142] | 37 |
| Late-Mar. 2018–Feb. 2019 H4 support | Public active-stock contrast | 0.923 | [0.557, 1.531] | 37 |

Notes: Every stock-flow comparison uses identical community support. Ratios below one indicate a lower exposed-relative after-before ratio in 2019 than in the identical 2018 calendar experiment; they do not imply a decline in exposed levels. Intervals use equation-specific school-clustered $t$ critical values and are not joint confidence sets for the stock-flow gap. The public waves are separate crawler snapshots rather than a listing-transition panel. On the 54-community November support, the stock/flow point-ratio gaps are 1.97 and 1.79; joint school-score equality tests give $p = 0.085$ and $p = 0.137$ using the geometric mean of the equation-specific CR1 finite-sample corrections. The smaller H4 support gives larger point gaps but only four exposed-school clusters. This is cross-crawler comparability sensitivity, not an identified execution hazard.

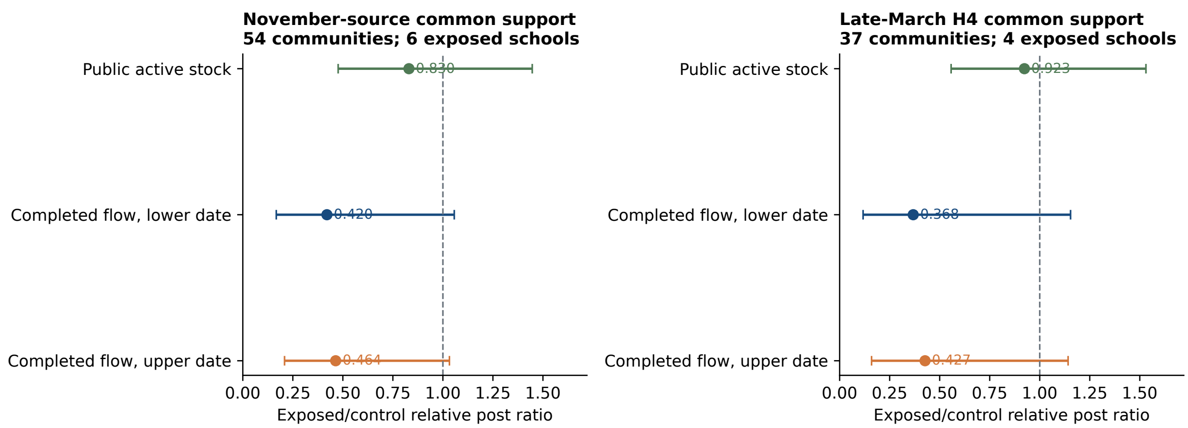


Figure A5: Completed flow and public active stock on identical supports

Notes: Intervals describe equation-specific uncertainty and do not form a joint confidence set for the stock-flow difference. Each comparison uses identical community support within data source.

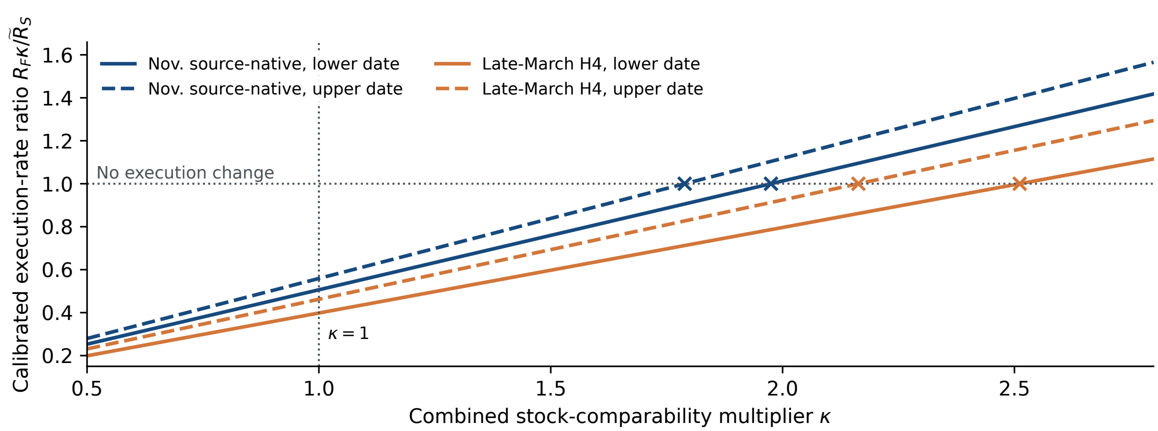


Figure A6: Comparability sensitivity for the execution interpretation

Notes: Curves apply the stock-flow identity to point estimates on identical supports. Crosses mark the comparability multiplier required for no execution change. The figure is not a confidence set and assigns no probability to $\kappa$.

## E Reproducibility and Scope of Claims

Hashed inputs, assignment samples, and machine-readable result files are recorded in the analysis protocols. A core exhibit builder regenerates the underlying cutoff estimates and figures, while separate locked routines generate assignment-stability, road-network, persistence, pre-announcement value, and selected-attention audits. The submission tables are assembled from machine-readable outputs produced by these documented modules. Independent routines reproduce sample construction, reuse weights, regressions, wild-cluster and CR2 inference, aggregation, and leave-one-out estimates. For the four headline and exact-notice asinh coefficients, a standalone audit reproduces the archived 9,999-draw Monte Carlo counts and then exhausts all $2^{21}$ assigned-school Rademacher sign vectors under the same restricted-null, fully studentized procedure. The reported exhaustive values are exact for that bootstrap sign distribution, not assignment-based randomization $p$-values.

The replication folder accompanying this appendix contains the full assignment audit, frozen matched frame and pairs, machine-readable headline and long-horizon results, selected-completer summaries, the frozen selected-duration-tail protocol and script, and the incidence-promotion and displacement-diagnostic scripts with their machine-readable results. The main transaction input is proprietary and cannot be redistributed without provider permission. The public notice images, source URLs, and their hashes remain separately auditable.

The interpretation used throughout the paper is:

1. **Identified under parallel differential seasonality:** a cutoff-local exposed-versus-ordinary DDD in completed transaction flow, not an absolute exposed decline or districtwide contraction.
2. **Exactly accounted but selected:** entry, exit, and stock among eventual sales.
3. **Conditionally calibrated:** execution as a function of public stock and $\kappa$.
4. **Not identified:** full inventory, listing-level sale probability, withdrawal, relisting, buyer arrival, permanent trade destruction, or welfare.

## References


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